

Losing Access to Canton: Trade Shock and Social Conflict in Qing China

Zixin Wang*

September 30, 2026

Abstract. Does losing access to an established trade route stabilize or destabilize society? Economic theory is ambiguous. I examine the 1842 Treaty of Nanjing, which ended the Canton System and opened four new coastal treaty ports, collapsing demand for transport labor along the old inland routes. Using an original county-year panel of 427 counties from 1820 to 1860, I find that counties on the pre-1842 Canton routes experienced a large rise in rebellion probability after the treaty. Counties in the hinterland of Shanghai, which gained trade access, show no comparable rise. This asymmetry, together with effects concentrated in counties that served as transit hubs, points to an opportunity cost channel through displaced transport workers rather than rapacity. Disaggregating conflict by motive, the effect holds for both anti-social and anti-state violence, and route counties became substantially more susceptible to Taiping mobilization after 1851.

1 Introduction

Whether access to trade opportunities enhances or undermines social stability is one of the most actively debated questions in economics and political science. The theoretical relationship is ambiguous. Under the opportunity cost mechanism, trade raises incomes and provides employment, increasing the cost of participation in violence and thereby stabilizing society (Becker, 1968). Under the rapacity mechanism, trade generates contestable rents that rival groups compete to capture, fueling conflict rather than suppressing it (Grossman, 1991). Identifying which mechanism dominates is empirically challenging: trade and conflict are mutually determined, so observed correlations between market integration and social unrest may reflect the effect of conflict on trade as much as the reverse. In an era of rising

*Zixin Wang: Department of Economics, London School of Economics and Political Science (email: z.wang111@lse.ac.uk). I am profoundly grateful to my supervisors, Ronny Razin and Kent Deng, for their invaluable guidance. I also thank Tim Besley, Ryan Jablonski, Noam Yuchtman, Zhiwu Chen, Melanie Meng Xue, and participants at the 12th Symposium on Quantitative History and various other seminars for their helpful suggestions. We alone are responsible for any remaining errors.

trade protectionism and de-globalization, understanding the causal effect of losing access to established trade routes on local social stability is both a pressing empirical question and a matter of immediate policy relevance. This paper provides new causal evidence by exploiting a sharp, externally imposed institutional break, the 1842 Treaty of Nanjing, which abruptly disrupted the inland trade routes serving China’s Canton System and plausibly raised the propensity for rebellion among hundreds of thousands of displaced transport workers.

Between 1757 and 1842, the Qing court restricted all Western maritime trade to a single port: Guangzhou, known to Western merchants as Canton. Under the resulting Canton System, export commodities, primarily tea from Anhui and Fujian and raw silk from the Jiangnan delta, were carried southward through inland provinces along an established network of riverine and overland routes, converging on Guangzhou for export to Western markets. The operation of these routes sustained a vast transport labor force: historians estimate that at least one million porters and boatmen earned their livelihoods through trade-related transport along the Canton routes (Wu, 2018). The system was dismantled following China’s defeat in the First Opium War. The Treaty of Nanjing, signed in August 1842, compelled the Qing court to open four new coastal treaty ports (Xiamen, Fuzhou, Ningbo, and Shanghai) and abolished the commercial monopoly that had sustained the Canton trade. Because the new northern ports were geographically far closer to the interior producing regions, trade rapidly reallocated toward the coast. Shipping Anhui tea via Shanghai required approximately twenty days, compared with two months via Canton (Fortune, 1999). Canton’s share of national exports, which had stood near unity before 1842, collapsed within a decade, while Shanghai emerged as the dominant center of China’s overseas commerce. For the workers whose livelihoods had depended on the Canton route economy, this reallocation meant the sudden destruction of their primary source of income.

This historical context offers several features that make it well-suited for identifying the causal effect of trade disruption on social conflict. First, the shock was imposed by military defeat and diplomatic settlement. It was neither trade-oriented nor motivated by existing or anticipated rebellions, providing a plausibly exogenous shift in trade access (Jia, 2014). Second, and more importantly, the 1842 shock redistributed foreign trade geographically rather than eliminating it. The opening of the treaty ports drew overseas commerce away from the Canton corridor and toward new export channels and their surrounding hinterlands. This feature of the setting allows me to distinguish between counties that were negatively exposed to the decline of the pre-1842 Canton system and counties that were positively exposed to the expansion of the new treaty-port network. This distinction serves as a useful interpretive check. If the estimated effects are concentrated in counties that lost access to the old trade corridor rather than in those that gained access to the new one, the results more

plausibly reflect disruption to established trade access than a general response to the 1842 institutional shock. Third, the Qing imperial archives record rebellion events with sufficient geographic and temporal detail to construct a county-year panel spanning both the pre- and post-treaty periods. Fourth, because identification comes from comparisons between route and non-route counties within provinces, the design helps absorb province-level shocks and reduces concerns that the results are driven by differential exposure to the broader military and political consequences of the First Opium War.

I construct a novel county-year panel covering 427 counties across six provinces from 1820 to 1860, by extracting rebellion events from two official Qing archives: the *Qing Shilu* (Veritable Records of the Qing Emperors) and the *Zhupi Zouzhe* (palace memorials with vermilion rescripts) from the First Historical Archives of China. I identify rebellion-related entries through keyword search and manual verification, and further classify each event as either anti-social violence (directed against local communities, merchants, or property) or anti-state violence (directed against state authority). To measure exposure to the 1842 trade shock, I digitize the pre-1842 inland routes to Canton from three historical sources and classify a county as treated if it lies on any of these route segments. The baseline analysis focuses on the 277 counties in the four provinces traversed by the inland routes (Anhui, Jiangxi, Fujian, and Guangdong), of which 51 are classified as treated and 226 serve as controls. The remaining 150 counties in Jiangsu and Zhejiang, positively exposed through their proximity to the new treaty ports, serve as a placebo sample examined in Section A.

I estimate a two-way fixed effects difference-in-differences (DID) specification that compares changes in rebellion probability between treated and control counties before and after 1842. I choose 1842, the year the Treaty of Nanjing was signed, as the treatment date, and I verify that this choice is consistent with the timing of changes observed in the data. In the preferred specification, with county and province-by-year fixed effects and the full set of controls, the annual probability of rebellion increased by approximately 10.9 percentage points in counties along the Canton routes after 1842, equivalent to about 77 percent of the full-sample mean of 0.141. The estimate is stable across specifications and remains robust when the binary rebellion indicator is replaced with a population-normalized measure constructed via an inverse hyperbolic sine transformation of per-capita rebellion counts.

I extend the baseline analysis in several directions. To characterize heterogeneity in treatment intensity, I construct a continuous exposure measure based on the pre-1842 density of market towns in each county, following the approach of [Cao and Chen \(2022\)](#). The treatment effect does not vary with market town density across the treated group, indicating that exposure is determined by corridor location rather than by the intensity of local commerce. I also distinguish transport-exposed counties, whose primary function in the Canton

network was to facilitate the movement of goods, from production-exposed counties, whose economies centered on commodity cultivation. The estimated effects are larger for the former group, though the difference between the two is only marginally significant. Spillover effects extend to off-route counties within approximately 100 km of the historical routes and attenuate sharply with distance, consistent with local labor market spillovers from displaced workers. Disaggregating conflict by motive, the effect is present for both anti-social and anti-state violence, indicating that the trade shock raised conflict broadly rather than along a single dimension. I further show that counties exposed to the Canton trade shock were more than twice as likely to experience Taiping-related activity after 1851 and faced a substantially higher conditional hazard of first Taiping entry, directly linking the 1842 trade disruption to the origins of one of the deadliest civil conflicts in recorded history.

One central concern for identification is that counties on the Canton routes may have been differentially affected by other consequences of the 1842 institutional break beyond the trade shock itself. These include the military disruption of the First Opium War and changes in Qing administrative priorities during the treaty-port era. I address this concern in several ways. Province-by-year fixed effects absorb any province-level time-varying shocks, including differential provincial exposure to the Opium War’s military theater. The event-study estimates show no evidence of differential pre-trends between treated and control counties prior to 1842, supporting the identifying assumption. The most direct threat to interpretation is the possibility that the results reflect a general response to the 1842 shock rather than the trade disruption specifically. This concern is addressed by the null result for positively exposed counties: if the effect reflected political trauma, institutional disruption, or military dislocation common to all counties in the sample, one would expect counties near the new treaty ports to also show elevated conflict. The complete absence of any significant effect for these counties indicates that the results are specific to the disruption of the Canton route economy. A placebo test on the courier road network, reported in Section 5, further supports the causal interpretation.

The evidence is most consistent with an opportunity cost channel operating through the displacement of transport labor. As the Canton route economy collapsed, the porters, boatmen, and other workers who had depended on the through traffic of the corridor lost their livelihoods, and in a largely pre-industrial rural economy with few alternative opportunities, the cost of participating in collective violence fell sharply. Several features of the results support this interpretation. The null result for positively exposed counties rules out a rapacity mechanism. Under rapacity, the contestable rents created by trade access would raise conflict in the counties that gained access to the new treaty-port trade, while the counties that lost the Canton trade would, if anything, experience a decline in conflict. The data show

the opposite pattern: conflict rose sharply in counties that lost access and did not change in those that gained it. The effect is concentrated in transport-exposed counties rather than commodity-producing ones, consistent in direction with a labor displacement story, since transit workers faced a near-complete loss of demand for their services while producers could reallocate land and labor over time. The effect does not scale with commercial density within the treated group but is instead determined by corridor location itself, consistent with displaced livelihoods tied to the through traffic of the routes rather than to the breadth of local commerce. Finally, treated counties experienced a substantially higher incidence of secret society foundings after 1842, corroborating the recruitment channel through which displaced transport workers were drawn into organized resistance.

This paper contributes to three strands of literature. The first is the large and influential literature on economic shocks and civil conflict ([Miguel, Satyanath, and Sergenti, 2004](#); [Dube and Vargas, 2013](#); [Bazzi and Blattman, 2014](#)). Most of this literature studies transitory income, price, or weather shocks, and findings are mixed. [Dube and Vargas \(2013\)](#) find a destabilizing effect of commodity price shocks in Colombia, while [Bazzi and Blattman \(2014\)](#) find a null effect in a broader cross-country sample. I contribute by studying a long-term structural shock to trade routes that permanently destroyed the employment base of a large transport workforce, rather than temporarily affecting agricultural incomes. The permanent nature of the disruption, combined with the absence of alternative livelihoods for the affected workers, may account for the large and persistent effects I document.

The second and most closely related strand is the literature on trade access and social stability, including work on transportation infrastructure and conflict ([Dell, Feigenberg, and Teshima, 2019](#); [Cao and Chen, 2022](#)). This paper is most directly related to [Cao and Chen \(2022\)](#), who study the social consequences of the abandonment of China’s Grand Canal, the world’s largest artificial waterway, for rebelliousness in nineteenth-century North China. The two papers share a broad research design. Both exploit variation in county-level exposure to a historically established Chinese trade route and use Qing archival data on rebellion in a DID framework. The two studies nonetheless differ in ways that are substantively important. [Cao and Chen \(2022\)](#) study a disruption to internal, domestic trade, since the Grand Canal transported tribute grain and domestic commodities between northern and southern China, whereas I study the disruption of international trade, in which the Canton routes connected China’s interior producing regions to global export markets. The nature of the shocks also differs. [Cao and Chen \(2022\)](#) analyze a gradual administrative abandonment that began in 1826 and unfolded over several decades, while the Treaty of Nanjing represents a sharp, externally imposed institutional break concentrated in a single year. Most fundamentally, the mechanism of disruption differs across the two settings. In [Cao and Chen \(2022\)](#), the

Grand Canal gradually lost navigability, disrupting trade access for counties located along the canal. In my setting, by contrast, the old Canton routes were not physically destroyed; instead, they lost commercial importance as overseas trade was spatially reallocated toward coastal shipping and the newly opened treaty ports after 1842. This feature generates a two-sided comparison between counties negatively exposed to the decline of the old Canton-centered trade corridor and counties positively exposed to the expansion of the new treaty-port network. That comparison is a distinctive empirical feature of my design. In particular, the absence of a comparable increase in rebellion among positively exposed counties is more consistent with an opportunity-cost mechanism than with a rapacity-based interpretation. Finally, whereas the Taiping Rebellion enters [Cao and Chen \(2022\)](#) mainly as a confounding historical event to be addressed, my paper brings the relationship between the 1842 trade shock and subsequent Taiping susceptibility to the center of the empirical analysis.

The third strand is the literature on social unrest in nineteenth-century China ([Bai and Kung, 2011](#); [Kung and Ma, 2014](#); [Bai and Jia, 2016](#)). Existing work in this literature emphasizes the roles of cultural norms, climate shocks, and political institutions in driving the chronic disorder that afflicted the late Qing Dynasty. I contribute by showing that an international trade shock, caused by military defeat and the unequal treaty of 1842, had large and lasting effects on domestic social order, first increasing local disorder and later making affected counties more vulnerable to Taiping mobilization. The findings suggest that the origins of China’s nineteenth-century social crisis were not solely internal but were shaped in important ways by the country’s violent integration into the global trading order.

The remainder of this paper is organized as follows. Section 2 provides historical background on the Canton System and the opening of the treaty ports after 1842. Section 3 describes the data. Section 4 presents the empirical strategy and results, including baseline estimates, treatment intensity analysis, and the link to the Taiping Rebellion. Section 5 presents robustness checks. Section 6 discusses potential mechanisms. Section 7 concludes.

2 Historical Background

2.1 The Canton System before 1842

From 1757 until the signing of the Treaty of Nanjing in 1842, the Qing Court confined all Western maritime trade to a single port: Guangzhou, known to Western merchants as Canton. Under the Canton System, foreign merchants could live and trade only in a designated area outside Guangzhou, and all trade had to go through a licensed group of Chinese merchant houses known as the Thirteen Hongs ([Jia, 2014](#)). This institutional arrangement

concentrated the entire flow of China’s overseas trade, dominated by tea and silk, through a single geographic node for nearly a century.

Because Guangzhou lies at the southern periphery of the country, far from the major centers of commodity production, export goods had to travel considerable distances before reaching the coast. Anhui province was the primary source of green tea, the Wuyi Mountains of Fujian supplied the black teas most prized by European buyers, and the Jiangnan delta, spanning parts of Jiangsu and Zhejiang, specialized in raw silk (Zhang, 2001; Yao, 1962). These commodities were transported southward through a network of mountain paths and navigable waterways that linked the interior provinces to the Pearl River delta. The operation of these routes depended almost entirely on manual labor. Porters carried goods over mountain passes where terrain precluded river transport, while boatmen navigated the river systems connecting the interior to Guangzhou. Historians estimate that at least one million people earned their livelihoods from trade-related transport employment along these routes (Wu, 2018). To capture exposure to this system, I digitize the pre-1842 inland routes to Canton from historical sources (Zhang, 2001; Yao, 1962; Fortune, 1999) and use them to identify counties whose local economies were tied to the Canton trade.

Figure 1 maps the pre-1842 inland trade routes reconstructed from historical sources. These routes form the geographic basis for the treatment classification used in the empirical analysis below.

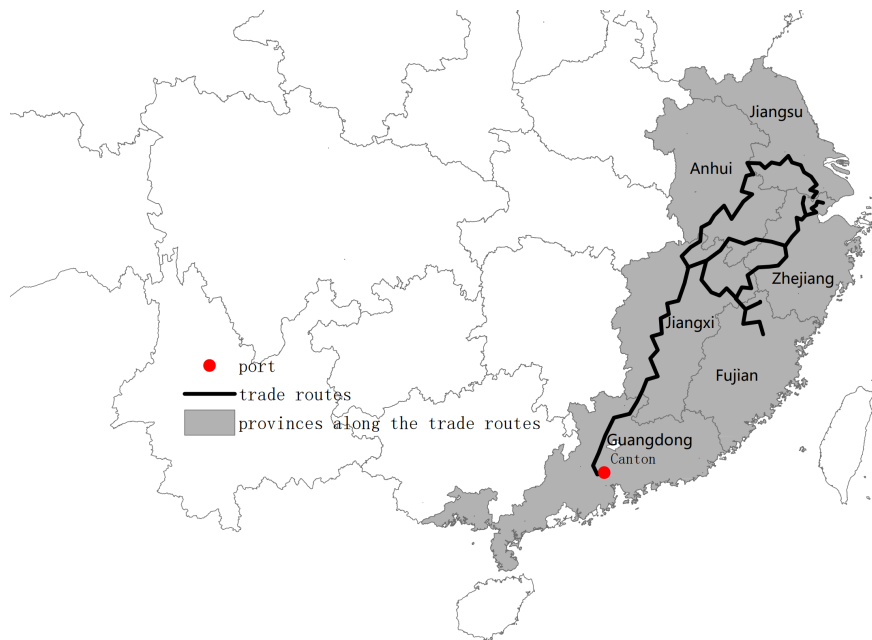


Figure 1: Pre-1842 Inland Trade Routes to Canton and Treaty Ports

Notes: The map shows the historical inland trade routes serving the Canton System across the six provinces in the sample (Jiangsu, Zhejiang, Anhui, Jiangxi, Fujian, and Guangdong), traced as thick black polylines. Red markers indicate the only port Guangzhou (Canton). Routes are digitized from [Zhang \(2001\)](#), [Yao \(1962\)](#), and [Fortune \(1999\)](#).

2.2 Opening of the Treaty Ports in 1842 and Trade Reallocation

China's defeat in the First Opium War (1839–1842) brought the Canton System to an abrupt end. The Treaty of Nanjing, signed in August 1842, compelled the Qing court to open four additional coastal cities to foreign trade: Xiamen, Fuzhou, Ningbo, and Shanghai. The monopoly of the Thirteen Hongs was simultaneously abolished, granting foreign merchants the right to trade directly at any of the five treaty ports. These institutional changes eliminated the geographic privilege that had sustained Canton's century-long position as China's exclusive gateway to Western commerce.

Trade reallocation away from Canton and toward the new treaty ports was rapid, driven primarily by transport costs. Prior to 1842, shipping Anhui green tea to foreign buyers required a journey of approximately two months via the inland routes to Canton; after the opening of Shanghai, the same goods could reach the coast in just over twenty days ([Fortune, 1999](#)). Given that transport costs had accounted for as much as one-third of total costs for tea, the cost advantage of the northern coastal ports was decisive.

As Figure 2 illustrates, the consequence was a swift and sustained collapse of Canton's

share of China’s total export trade.¹ Canton’s share, which had stood near unity before 1842, fell sharply in the years immediately following the treaty, while Shanghai emerged within a decade as the dominant center of China’s overseas commerce, handling the majority of both tea and silk exports by the early 1850s.

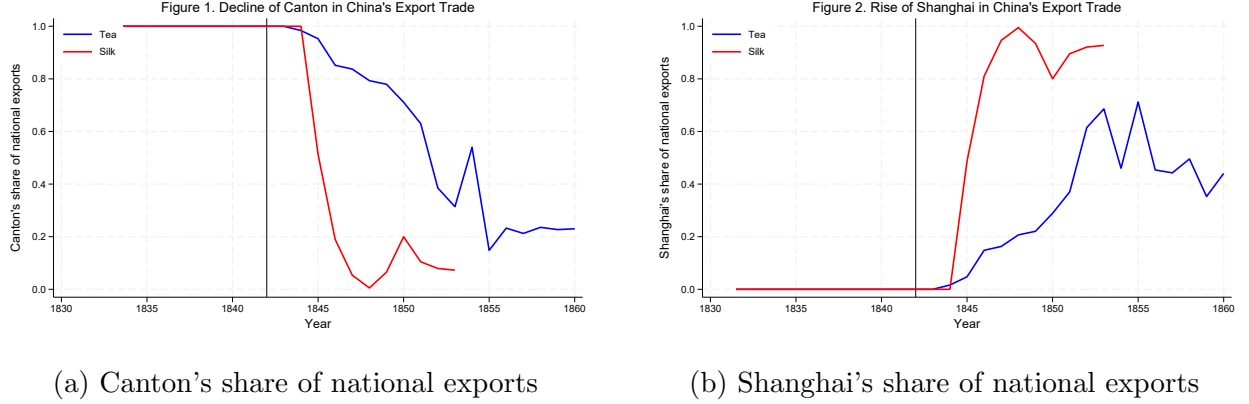


Figure 2: Decline of Canton and Rise of Shanghai in China’s Export Trade, 1830–1860

Notes: Each panel plots the share of China’s national exports handled by the indicated port for tea (red) and silk (blue), the two principal commodities of the Canton trade. The vertical dashed line marks 1842, the year of the Treaty of Nanjing. Panel (a) shows the rapid decline in Canton’s export share following the treaty. Panel (b) shows the corresponding rise of Shanghai as the dominant center of China’s overseas trade by the early 1850s. Source: Yao (1962).

For the workers whose livelihoods depended on the old inland routes, this trade reallocation represented a sudden and severe contraction in demand for their services. As commerce shifted toward coastal shipping, the need for overland and riverine transport labor collapsed. Wakeman (1966) observes that the resulting pool of displaced porters and boatmen became ready recruits for secret societies and armed resistance. The remainder of this paper asks whether this disruption translated into a measurable increase in local social conflict, using county-level variation in exposure to the collapsed Canton trade network as the basis for identification.

3 Data

3.1 Rebellions

The main outcome is a binary indicator equal to one if at least one rebellion event occurred in county c in year t . I construct a novel county-year panel covering the period

¹The analysis focuses on tea and silk, as these two commodities accounted for the bulk of China’s total export value during the mid-nineteenth century.

1820 to 1860 by extracting rebellion events from two official Qing imperial sources: the *Qing Shilu* (Veritable Records of the Qing Emperors) and the *Zhupi Zouzhe* (palace memorials with vermilion rescripts) from the First Historical Archives of China. For the years 1820 to 1829, rebellion records are drawn exclusively from the *Qing Shilu*; from 1830 onward, the panel combines the *Qing Shilu* with the *Zhupi Zouzhe*. These records constitute the official archive of imperial edicts and memorials reporting events of national significance, and are digitized on the Chinese Text Project and Shidian Guji platforms. Chinese historians regard the Qing Shilu as a uniquely authoritative source that offers the most reliable and comprehensive record of social unrest during the Qing Dynasty Yang (1975). To identify rebellion-related entries, I perform a keyword search using a set of terms common in Qing administrative language for civil unrest: *fei* (bandits), with frequent compounds including *tu fei* (local brigands), *jiao fei* (bandits linked to religious organizations), *yue fei* (Guangdong bandits), and *hui fei* (secret-society bandits); *zi shi* (to cause disturbances); *ju zhong* (to gather a mob); and *ni* (insurgent). All candidate entries identified by the keyword search are then manually reviewed in their original textual context to verify relevance.

I further classify each rebellion event by its primary target, following the distinction between anti-social and anti-state violence. *Anti-social violence* refers to collective violence directed primarily against local communities, merchants, property, or commercial resources rather than state authority. Typical cases include robbery and looting targeting merchants or local households, clan feuds involving the destruction of property, and armed clashes over local economic resources. *Anti-state violence* refers to collective violence directed primarily against the apparatus of the Qing state. Typical cases include targeted attacks on *yamen* (government offices), government officials, or military outposts; jail-breaking and armed resistance to arrest; and sectarian or secret-society uprisings explicitly aimed at challenging state authority. This distinction allows me to examine whether the trade shock altered not only the frequency of conflict but also its political character over time.

3.2 Treatment

Binary treatment. The baseline treatment variable $Treated_c$ is a binary indicator equal to one if county c lies on any of the pre-1842 inland trade routes to Canton. I identify these routes by digitizing three historical sources: Zhang (2001), Yao (1962), and Fortune (1999). A county is classified as treated if its administrative boundary intersects one of these digitized route segments. Of the 277 baseline counties in the four provinces (Anhui, Jiangxi, Fujian, Guangdong) covered by the sample, 57 are classified as treated.

Transport and production exposure. The binary treatment pools two economically distinct forms of exposure to the Canton trade network. *Transport-exposed* counties are those whose primary role in the network was to facilitate the movement of goods, serving as transit hubs for porters, boatmen, and commercial intermediaries. *Production-exposed* counties are those whose economies centered on the cultivation or processing of the principal export commodities, chiefly tea and raw silk. Because the trade shock disrupted both the demand for transport labor and the market access of commodity producers, these two groups may have faced different magnitudes of economic disruption. I construct separate indicators for transport and production exposure to test for heterogeneity in the treatment effect.

Continuous exposure. The binary indicator records only whether a county lay on the Canton routes, not how severely trade through those routes contracted after 1842. To capture the magnitude of the disruption, I construct a continuous exposure measure, $\text{TransitLoss}_{c,t} = \text{Treated}_c \times (1 - s_{\text{Canton},t})$, where $s_{\text{Canton},t}$ is Canton’s share of China’s overseas exports in year t , taken from Yao (1962).² The measure is zero for all control counties and for treated counties before 1842, and rises continuously as Canton’s export share is reallocated toward the new treaty ports. To distinguish between tea and silk, which followed different export routes and experienced declines at different times, I construct commodity specific flow loss measures, $\text{FLR}_{c,t}^k = \mathbf{1}\{k \text{ passes through the corridor containing county } c\} \times (1 - s_{\text{Canton},t}^k)$ for $k \in \{\text{tea, silk}\}$. Each treated county is assigned to the commodity or commodities historically transported along its segment of the route network.

Economic intensity. To capture variation in the depth of market integration within the treated group, I construct two continuous intensity measures following Cao and Chen (2022): the pre-1842 density of market towns,

$$\text{TownDensity}_{c,1820} = \frac{\text{MarketTowns}_{c,1820}}{\text{Area}_c} \times 10,000,$$

measured in towns per 10,000 km², and the share of a county’s 1820 market towns located within ten kilometers of the Canton routes. Both measures equal zero for counties off the corridor, and the intensity variable enters the analysis interacted with treatment, $\text{Intensity}_c = \mathbf{1}\{\text{Treated}_c = 1\} \times \text{TownDensity}_{c,1820}$. These measures distinguish counties more or less deeply embedded in the Canton trade network, allowing a test of whether the treatment effect scales with the depth of commercial integration or is instead determined by corridor location alone.

²Tea and silk were the two principal commodities of China’s overseas trade in this period; $s_{\text{Canton},t}$ is the volume-weighted combination of Canton’s tea and silk export shares.

3.3 Control Variables

The analysis controls for a rich set of pre-determined county characteristics that may independently predict both route placement and conflict propensity. *Geography.* I include each county’s land area (in km²) and a terrain ruggedness index (Nunn and Puga, 2012), both constructed from county-level shapefiles provided by the Harvard Yenching Institute and the Fudan Center for Historical Geography (Harvard Yenching Institute and Fudan Center, 2007). Rugged terrain raises the cost of state suppression and may independently foster conflict; it may also have shaped the placement of trade routes.

Demography. I account for historical population using county level estimates for 1820 from Zheng and Wu (2026). This source provides direct population estimates for counties in historical China and validates them against prefecture level figures reported in *The Population History of China* (Cao, 2001) and county household records compiled by Liang (1980).

Agriculture. I include indices of suitability for wetland rice and wheat cultivation from IIASA and FAO (2012) as proxies for each county’s agricultural productivity and degree of commercial orientation. I also control for crop failure using a county year indicator constructed from the *Qing Shilu* through the same keyword search and manual verification procedure described above.

Climate. To capture annual fluctuations in climatic conditions, I include a county level temperature anomaly based on the historical reconstructions of Mann et al. (2009a;b). I also control for yearly flood and drought events using indicators derived from the gridded hydroclimate dataset of Hao et al. (2023).

Table 3.3 presents the means and standard deviations for treated and control counties, together with the difference in means and its standard error. The two groups are broadly similar in their geographic characteristics, particularly in terms of land area and terrain ruggedness, and they exhibit almost identical average temperature anomalies. Treated counties nevertheless experience floods, droughts, and crop failures somewhat more frequently. This pattern may reflect the location of the Canton trade routes along river valleys that were vulnerable to seasonal flooding. Treated counties also tend to have larger populations and greater commercial activity, as indicated by their higher density of market towns. These differences observed before treatment motivate the use of province by year fixed effects in the preferred specifications to account for time varying shocks at the provincial level, together with the full set of time varying controls described above.

Table 1: Comparisons between treated and control counties

	Treated	Control	Difference
	(1)	(2)	(1)-(2)
County area (km ²)	2157.29 (1048.59)	2231.37 (1059.55)	-74.08 [157.14]
Ruggedness Index	144.239 (93.095)	167.712 (95.930)	-23.473 [14.173]
Temperature Anomaly	0.388 (0.046)	0.358 (0.052)	0.030 [0.008]
Frequency of flooding	0.464 (0.047)	0.458 (0.081)	0.007 [0.011]
Frequency of droughts	0.192 (0.061)	0.225 (0.080)	-0.033 [0.011]
Population in 1820	491,543 (509,968)	298,460 (240,536)	193,083 [46,946]
Suitable for wheat	2290.842 (808.991)	2449.840 (1017.973)	-158.998 [145.510]
Suitable for wetland rice	3035.989 (1070.866)	2657.418 (1155.710)	378.572 [169.275]
Crop failure	0.162 (0.182)	0.082 (0.142)	0.079 [0.022]
Market towns	4.035 (2.507)	3.968 (2.832)	0.067 [0.412]
Observations	57	220	277

Note: Treated counties lie on the inland trade routes to Canton that were active before 1842; control counties are all other counties in the same four provinces (Anhui, Jiangxi, Fujian, and Guangdong). Standard deviations in parentheses; standard errors of the difference in brackets. Time-varying variables (Temperature Anomaly, Frequency of flooding, Frequency of droughts, and Crop failure) are county-level averages over 1820 to 1860 (Temperature Anomaly and Crop failure, unavailable for 1820 to 1829, average over 1830 to 1860). Population is county-level population in 1820 from [Zheng and Wu \(2026\)](#); it is available for 274 of the 277 counties.

3.4 Suggestive Evidence

Before turning to the formal empirical strategy, Figure 3 offers a descriptive overview of the joint geography of the trade shock and the conflict outcome. The map overlays the pre-1842 inland routes to Canton with the spatial distribution of changes in the annual rebellion rate across counties. For each county, this change is defined as the average number of rebellions per year in the post-treaty period (1843 to 1860) minus the average number per year in the pre-treaty period (1830 to 1841), with 1842 excluded as the year of the shock. Expressing the change as a per-year rate makes the two periods comparable despite their different lengths. The counties with the largest increases in the annual rebellion rate, shown in darker shades of gray, are disproportionately concentrated along or near the historical Canton routes rather than distributed uniformly across the four provinces of Anhui, Jiangxi, Fujian, and Guangdong. These patterns motivate the difference-in-differences design, which exploits cross-county variation in pre-1842 route exposure to identify the causal effect of trade-access loss on local unrest.

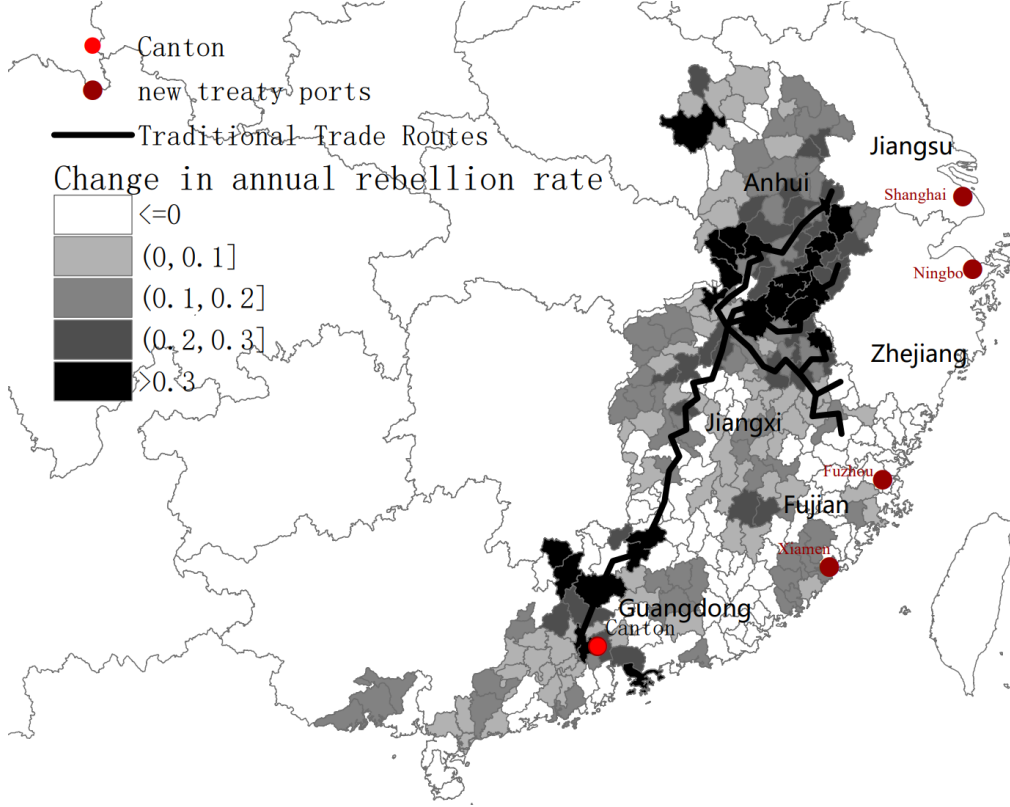


Figure 3: Spatial Distribution of Changes in the Annual Rebellion Rate around 1842

Notes: The map overlays the pre-1842 inland trade routes to Canton (black lines) and the treaty ports opened after the Treaty of Nanjing (red points), with Canton itself marked separately as the sole port open to foreign trade before 1842. Shaded counties show the change in the annual rebellion rate, defined as average rebellions per year in the post-treaty period (1843 to 1860) minus average rebellions per year in the pre-treaty period (1830 to 1841), with 1842 excluded. The change is measured in incidents per year and binned into five categories. Darker shades indicate larger increases in the annual rebellion rate. Counties with no increase or a decrease are shown in white.

4 Empirical strategy and Results

4.1 Empirical Strategy

To identify the causal effect of trade-access loss on local social conflict, I exploit county-level variation in exposure to the pre-1842 inland Canton trade routes and estimate a two-way fixed effects difference-in-differences specification:

$$Y_{ct} = \beta(\text{Treated}_c \times \text{Post}_t) + \lambda_c + \delta_t + X_{ct}\Gamma + \varepsilon_{ct}, \quad (1)$$

where Y_{ct} is an indicator equal to one if at least one rebellion occurred in county c in year t ; Treated_c equals one if county c lies on the pre-1842 inland routes to Canton; Post_t equals one for years 1842

and after; λ_c denotes county fixed effects; and δ_t denotes year fixed effects. The county fixed effects absorb all time-invariant differences across counties, including baseline conflict propensity and geography. The year fixed effects absorb common shocks to all counties in a given year. The vector X_{ct} includes the time-varying controls described in Section 3: crop failure, temperature anomaly, drought, and flood. Standard errors are clustered at the county level throughout.

The coefficient of interest, β , identifies the average change in rebellion incidence in treated counties relative to control counties following the 1842 trade shock, under the assumption that, absent the shock, treated and control counties would have followed parallel trends in conflict. The preferred specifications replace year fixed effects with province-by-year fixed effects ($\varphi_{p(c),t}$), which absorb any province-level time-varying confounders, such as differential responses to the First Opium War or province-specific administrative shocks, and thereby sharpen identification to within-province comparisons over time.

Figure 4 illustrates the two sides of the 1842 trade reallocation. The left panel maps the counties that were negatively exposed to the shock, namely those whose pre-1842 role in the Canton trade network was disrupted by the opening of the treaty ports. The right panel maps the counties that were positively exposed, namely those in the coastal hinterlands of the new treaty ports that stood to gain from the reallocation of overseas trade. Comparing conflict dynamics across these two groups provides a test for whether the estimated effects reflect trade disruption specifically rather than a more general response to the 1842 institutional shock.

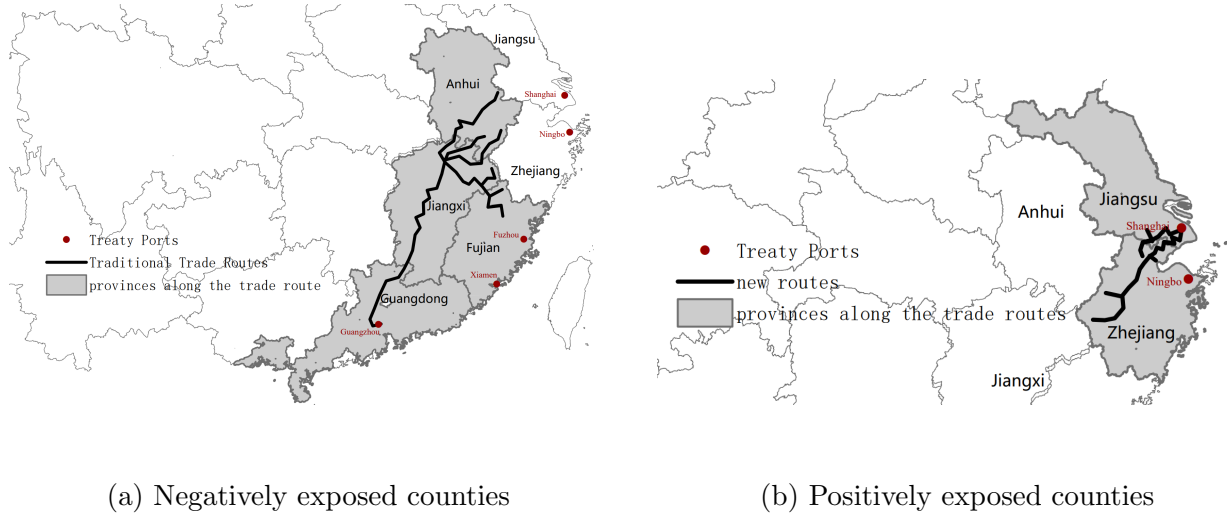


Figure 4: Negatively and Positively Exposed Counties

Notes: Panel (a) shows counties classified as negatively exposed to the 1842 trade shock, those lying on pre-1842 inland trade routes to Canton (shaded gray), alongside the traditional routes (black lines) and new treaty ports (red points). Panel (b) shows counties classified as positively exposed, those lying in the hinterlands of the new treaty ports opened after 1842.

To assess the validity of the parallel trends assumption and to characterize the dynamics of

the treatment effect, I estimate the following event-study specification:

$$Y_{ct} = \sum_{k \neq 0} \beta_k \left(\text{Treated}_c \times \mathbf{1}\{t \in \mathcal{B}_k\} \right) + \lambda_c + \varphi_{p(c),t} + X_{ct}\Gamma + \varepsilon_{ct}, \quad (2)$$

where $\mathcal{B}_k$ denotes two-year time bins centered on the reference period 1841–1842, and the omitted category is the bin immediately preceding the treaty (1841–1842). The coefficients β_k for $k < 0$ test for pre-existing differential trends, while coefficients for $k > 0$ trace the post-shock trajectory of the treatment effect.

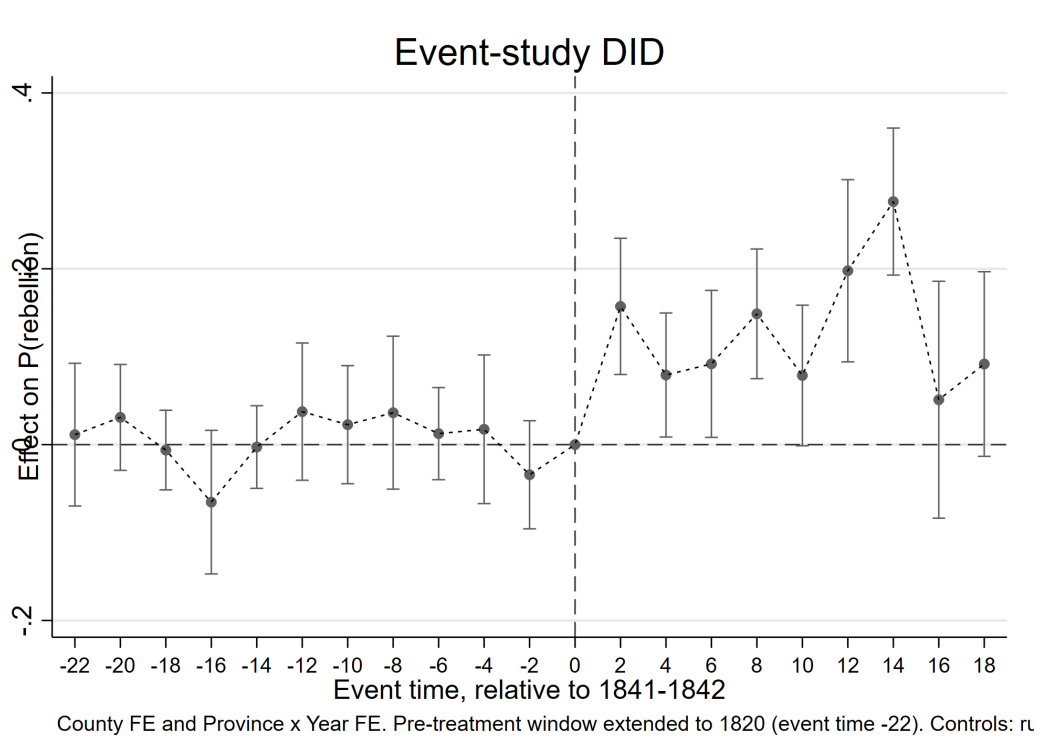


Figure 5: Event-Study Estimates: Trade Shock and Rebellions

Notes: The figure plots event-study coefficients $\hat{\beta}_k$ from equation (2), with 95 percent confidence intervals. The sample covers 277 counties in four provinces from 1820 to 1860. The omitted reference period is 1841 to 1842. Time bins are two years wide. Specifications include county fixed effects, province-by-year fixed effects, and the full set of time-varying controls. Standard errors are clustered at the county level.

The pre-treatment coefficients are small and statistically indistinguishable from zero, and an F -test cannot reject the null that they are jointly equal to zero, supporting the parallel trends assumption. After the treaty, the coefficients rise progressively, indicating that the effect of the trade shock emerged gradually rather than all at once, a pattern consistent with the slow adjustment of trade flows documented in Figure 2.

4.2 Baseline Estimates

Negative trade exposure. Table 2 reports estimates of equation (1) for the full sample of 277 counties between 1820 and 1860. Across all six specifications, the coefficient on $Treated \times Post-1842$ is positive, precisely estimated, and stable.

Column (1) includes county and year fixed effects without additional controls, yielding an estimate of 0.1364. Columns (2) through (6) replace year fixed effects with province-by-year fixed effects and sequentially add time-varying controls for crop failure, climate anomaly, drought, and flood. The estimate remains stable across these specifications. In the preferred specification (Column 6), which includes province-by-year fixed effects and the full set of controls, exposure to the Canton trade shock raised the annual probability of rebellion by approximately 10.9 percentage points. Relative to the full-sample mean rebellion rate of 0.141, this represents an increase of about 77 percent. The stability of the coefficient across specifications suggests that the estimated effect is not driven by observable time-varying confounders.

Table 2: Trade Shock and Rebellions: Baseline Estimates

	Rebellion occurrence (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Treated $\times$ Post-1842	0.1364*** (0.0184) [0.0256]	0.1169*** (0.0187) [0.0224]	0.1169*** (0.0186) [0.0224]	0.1176*** (0.0186) [0.0224]	0.1080*** (0.0185) [0.0217]	0.1093*** (0.0184) [0.0218]
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Prov. $\times$ Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	No	No	Yes
Geography controls	No	No	No	Yes	No	Yes
March-route control	No	No	No	No	Yes	Yes
Mean of dependent variable	0.1412	0.1412	0.1412	0.1412	0.1412	0.1412
Number of counties	277	277	277	277	277	277
Observations	11,356	11,356	11,356	11,356	11,356	11,356
Adjusted R^2	0.201	0.328	0.328	0.328	0.329	0.329

Notes: The dependent variable is an indicator equal to one if at least one rebellion occurred in county c in year t . Treated $\times$ Post-1842 equals one for counties on the pre-1842 inland routes to Canton in years ≥ 1842 . All columns include county fixed effects. Climate controls comprise drought and flood indicators. The geography control is terrain ruggedness $\times$ Post-1842. The march-route control is the distance from each county centroid to the Taiping 1851 to 1853 march route, interacted with Post-1851. Standard errors clustered at the county level are in parentheses; Conley spatial-HAC standard errors with a 500 km cutoff and 10-year serial correlation are in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Positive trade exposure. As a placebo-style test, I estimate equation (1) for the counties positively exposed to the new treaty ports, those whose geographic position gave them potential

access to the reallocated overseas trade. If the estimated effect in Table 2 reflects the disruption of the Canton network specifically, rather than a general response to the 1842 institutional shock, the coefficient for positively exposed counties should be small and insignificant. The results, reported in Table A2 in the Appendix, confirm this prediction: the coefficient on $\text{Treated} \times \text{Post-1842}$ ranges from about -0.016 to 0.002 across specifications and is statistically indistinguishable from zero in all cases, against a pre-1842 mean of 0.046 (full-sample mean 0.092). The asymmetry between negative and positive exposure is inconsistent with a rapacity mechanism. Under rapacity, the rents created by trade access would raise conflict in the counties that gained access to the new treaty-port trade, yet these counties show no increase, while the counties that lost access show a large one. It is instead consistent with an opportunity-cost interpretation, in which the relevant channel is the destruction of livelihoods rather than the creation of rents.

Conflict by motive: anti-social and anti-state violence. To examine whether the trade shock raised conflict of a particular character or across the board, I re-estimate equation (1) separately for anti-social and anti-state violence, using as the outcome an indicator for each type of event.

Table 3 reports the results for anti-social violence. The coefficient on $\text{Treated} \times \text{Post-1842}$ is approximately 0.037 and significant across specifications, against a sample mean of 0.047 . Anti-social violence comprises economically motivated conflict, including robbery, looting, and local armed clashes, the forms of disorder most directly associated with displaced transport workers facing acute income loss.

Table 3: Trade Shock and Rebellions: Anti-Social Violence

	Economic-motive rebellions					
	(1)	(2)	(3)	(4)	(5)	(6)
Treated $\times$ Post-1842	0.0419*** (0.0117)	0.0514*** (0.0114)	0.0512*** (0.0114)	0.0517*** (0.0114)	0.0489*** (0.0114)	0.0487*** (0.0114)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Province $\times$ Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	No	No	Yes
Geography controls	No	No	No	Yes	No	Yes
March-route control	No	No	No	No	Yes	Yes
Mean of dependent variable	0.053	0.053	0.053	0.053	0.053	0.053
Number of counties	277	277	277	277	277	277
Observations	11357	11357	11357	11357	11357	11357
Adjusted R^2	0.075	0.134	0.134	0.134	0.134	0.135

Notes: The dependent variable is an indicator equal to one if at least one anti-social rebellion occurred in county c in year t . Anti-social violence is defined as collective violence directed primarily against local communities, merchants, or commercial property. All columns include county fixed effects. Column (1) includes year fixed effects; Columns (2)–(5) include province $\times$ year fixed effects. Standard errors, clustered at the county level, in parentheses.

Table 4 reports the corresponding results for anti-state violence, defined as collective violence directed against state authority. The coefficient on Treated $\times$ Post-1842 is approximately 0.041 and significant across specifications, against a sample mean of 0.112. That the effect appears for both anti-social and anti-state violence indicates that the trade shock raised conflict broadly rather than along a single dimension. The anti-state response is of particular interest because organized resistance against the state is the form of conflict most closely connected to the large-scale insurrection of the 1850s, a connection I examine directly in Section 4.4.

Table 4: Trade Shock and Rebellions: Anti-State Violence

	Political-motive rebellions					
	(1)	(2)	(3)	(4)	(5)	(6)
Treated $\times$ Post-1842	0.0940*** (0.0139)	0.0680*** (0.0143)	0.0682*** (0.0142)	0.0683*** (0.0143)	0.0630*** (0.0142)	0.0643*** (0.0141)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Province $\times$ Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	No	No	Yes
Geography controls	No	No	No	Yes	No	Yes
March-route control	No	No	No	No	Yes	Yes
Mean of dependent variable	0.093	0.093	0.093	0.093	0.093	0.093
Number of counties	277	277	277	277	277	277
Observations	11357	11357	11357	11357	11357	11357
Adjusted R^2	0.269	0.423	0.424	0.423	0.423	0.425

Notes: The dependent variable is an indicator equal to one if at least one anti-state rebellion occurred in county c in year t . Anti-state violence is defined as collective violence directed primarily against state authority, including attacks on government offices or officials, jail-breaking, and secret-society uprisings. All columns include county fixed effects. Column (1) includes year fixed effects; Columns (2)–(5) include province $\times$ year fixed effects. Standard errors, clustered at the county level, in parentheses.

4.3 Treatment Intensity

Transport versus production exposure. The binary treatment indicator conflates two economically distinct channels through which the Canton trade network shaped local economies: transport-dependent counties whose primary role was to facilitate the movement of goods along the routes, and production-dependent counties whose economies centered on the cultivation or processing of export commodities. To assess whether the treatment effect differs across these two groups, I estimate:

$$Y_{ct} = \beta_1 (\text{TransportTreated}_c \times \text{Post}_t) + \beta_2 (\text{ProductionTreated}_c \times \text{Post}_t) + \lambda_c + \varphi_{p(c),t} + X_{ct}\Gamma + \varepsilon_{ct}, \quad (3)$$

where $\text{TransportTreated}_c$ and $\text{ProductionTreated}_c$ are mutually exclusive indicators partitioning the treated group.

Table 5 reports the results. The coefficient on $\text{TransportTreated} \times \text{Post-1842}$ is approximately 0.114 to 0.141 and statistically significant at the 1 percent level across all specifications. The coefficient on $\text{ProductionTreated} \times \text{Post-1842}$ is smaller, ranging from about 0.064 to 0.117, and is

statistically significant at the 5 percent level or better in every specification. The point estimate for transport-exposed counties consistently exceeds that for production-exposed counties. This pattern is consistent with the view that the displacement of transport labor, which was the most direct consequence of route collapse, was a primary driver of unrest. The p -values for the equality of the two coefficients range from 0.10 to 0.54 across specifications and equal 0.10 in the preferred specification, so the two effects are at most marginally distinguishable with the current sample size. The results are best interpreted as evidence that the transport channel is the larger of the two, while both forms of exposure raise rebellion.

Table 5: Trade Shock and Rebellions: Transport-Exposed versus Production-Exposed Counties

	Rebellion occurrence (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Transport $\times$ Post-1842	0.1411*** (0.0204) [0.0276]	0.1224*** (0.0221) [0.0242]	0.1224*** (0.0221) [0.0242]	0.1305*** (0.0223) [0.0248]	0.1141*** (0.0219) [0.0236]	0.1221*** (0.0220) [0.0245]
Production $\times$ Post-1842	0.1170*** (0.0353) [0.0481]	0.0973*** (0.0261) [0.0452]	0.0974*** (0.0262) [0.0452]	0.0726*** (0.0267) [0.0440]	0.0860*** (0.0260) [0.0439]	0.0644** (0.0262) [0.0427]
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Prov. $\times$ Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	No	No	Yes
Geography controls	No	No	No	Yes	No	Yes
March-route control	No	No	No	No	Yes	Yes
p -value: Transport = Production	0.540	0.449	0.451	0.101	0.392	0.097
Mean of dependent variable	0.1412	0.1412	0.1412	0.1412	0.1412	0.1412
Number of counties	277	277	277	277	277	277
Observations	11,356	11,356	11,356	11,356	11,356	11,356
Adjusted R^2	0.201	0.328	0.328	0.328	0.329	0.330

Notes: The sample consists of 277 counties in four provinces (Anhui, Jiangxi, Fujian, and Guangdong) from 1820 to 1860 (40 Transport-exposed and 11 Production-exposed). The dependent variable is an indicator equal to one if at least one rebellion occurred in county c in year t . Transport-exposed equals one for treated counties whose primary function in the Canton trade network was to facilitate the movement of goods (transport hubs along the inland trade routes). Production-exposed equals one for treated counties whose economies were primarily oriented toward export commodity production (tea, silk). Post-1842 equals one in years 1842 and after; Post-1851 equals one in years 1851 and after. All columns include county fixed effects. Column (1) adds year fixed effects; Columns (2) to (6) replace year fixed effects with province $\times$ year fixed effects. Climate controls comprise indicators for drought and flood (time-varying). The geography control is terrain ruggedness interacted with Post-1842. The march-route control is the geodesic distance from each county centroid to the Taiping 1851 to 1853 military advance route, interacted with Post-1851. The row p -value: Transport = Production reports the p -value from an F-test of the equality of the two interaction coefficients. Standard errors clustered at the county level are in parentheses; Conley spatial-HAC standard errors with a 500 km spatial cutoff and 10-year serial correlation are in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Economic intensity. The binary treatment indicator masks variation in the depth of market integration within the set of treated counties. Following [Cao and Chen \(2022\)](#), I use two continuous measures of a county’s integration into the Canton trade network: the pre-1842 density of market towns (towns per 10,000 km²) and the share of a county’s 1820 market towns located within ten kilometers of the Canton trade routes. Both measures equal zero for counties off the corridor. Adapting equations (4) and (5) of [Cao and Chen \(2022\)](#), I estimate a linear intensity specification,

$$Y_{ct} = \beta(\text{Intensity}_c \times \text{Post}_t) + \lambda_c + \varphi_{p(c),t} + X_{ct}\Gamma + \varepsilon_{ct}, \quad (4)$$

and a more flexible group specification that splits treated counties into intensity terciles,

$$Y_{ct} = \sum_{k=1}^3 \beta_k(\mathbf{1}\{\text{IntensityGroup}_c = k\} \times \text{Post}_t) + \lambda_c + \varphi_{p(c),t} + X_{ct}\Gamma + \varepsilon_{ct}, \quad (5)$$

where terciles are defined among the 51 treated counties and control counties form the omitted group. If the trade shock operates with a dose response, the tercile coefficients should rise monotonically in k ; if exposure is instead determined by corridor location itself, the three coefficients should be uniform.

Table 6 reports the estimates. The linear specifications in columns (1) and (2) yield positive and significant coefficients for both intensity measures, and the route share result holds under alternative bandwidths of 5 and 20 kilometers. The tercile specifications in columns (3) and (4) reveal that these linear estimates do not reflect a gradient within treated counties. All six tercile coefficients are positive, significant, and tightly clustered, with no monotone pattern: the lowest and highest density terciles yield nearly identical estimates, and equality of the first and third terciles cannot be rejected for either measure. The significant linear coefficients therefore capture the contrast between treated and control counties rather than scaling within the treated group.

The flat tercile profile is informative about the mechanism. Counties on the corridor experience a comparable rise in rebellion regardless of their commercial density or the proximity of their market towns to the route, so exposure is well described by corridor location itself. This pattern accords with the transit labor channel, in which the displaced livelihoods of porters, boatmen, and innkeepers were tied to the through traffic of the corridor rather than to the breadth of local commerce. A mechanism operating through general commercial contraction would instead predict larger effects in more commercialized counties, which the tercile estimates do not show.

Continuous exposure and commodity composition. The transport-production and market-town-density results above characterize *which* treated counties responded most strongly. A complementary question is whether the effect scales with the *magnitude* of the trade disruption itself, rather than with route membership alone. The binary treatment indicator records only whether a county lay on the pre-1842 Canton routes; it does not register how severely trade through those routes actually contracted. To capture this, I construct a continuous exposure measure that

Table 6: Trade Shock and Rebellions: Treatment Intensity

	(1)	(2)	(3) TownDensity	(4) RouteTownShare
TownDensity $\times$ Post	0.0037*** (0.0009)			
RouteTownShare $\times$ Post		0.1353*** (0.0378)		
Low intensity $\times$ Post			0.0866*** (0.0310)	0.0699** (0.0284)
Middle intensity $\times$ Post			0.1062*** (0.0335)	0.1206*** (0.0274)
High intensity $\times$ Post			0.0885*** (0.0276)	0.0947*** (0.0351)
Climate controls	Yes	Yes	Yes	Yes
Geography $\times$ Post	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Province $\times$ Year FE	Yes	Yes	Yes	Yes
Observations	8587	8587	8587	8587
Within R-squared	0.0047	0.0048	0.0057	0.0059
p -value: low = high			0.958	0.544

Notes: Dependent variable is an indicator equal to one if at least one rebellion occurred in the county-year. Standard errors clustered by county in parentheses. TownDensity is the number of 1820 market towns per 10,000 km²; RouteTownShare is the share of a county's 1820 market towns within 10 km of the Canton trade routes; both equal zero for control counties. Columns (1) and (2) interact each measure linearly with Post-1842. Columns (3) and (4) split the measure indicated in the column header into low, middle, and high terciles among the 51 treated counties (TownDensity: 17/17/17 counties per group; RouteTownShare: 19/16/16 owing to mass points at low shares), with control counties as the omitted group. The reported p -value tests equality of the low and high tercile coefficients. Results are robust to alternative bandwidths in the linear route-share specification: 5 km ($b = 0.143$, $p = 0.025$) and 20 km ($b = 0.117$, $p < 0.001$). All columns include county fixed effects, province $\times$ year fixed effects, and the climate and geography controls described in the text. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

interacts route membership with the year-by-year collapse of Canton’s export share:

$$\text{TransitLoss}_{c,t} = \text{Treated}_c \times (1 - s_{\text{Canton},t}), \quad (6)$$

where $s_{\text{Canton},t}$ is Canton’s combined (tea and silk, volume-weighted) share of China’s overseas exports in year t , constructed from the export series in Yao (1962). By construction TransitLoss is zero for all control counties and for treated counties before 1842, and rises continuously thereafter as Canton’s share is reallocated toward the new treaty ports. It thus converts the static treatment dummy into a dose that tracks the pace of trade reallocation documented in Figure 2.

I estimate

$$y_{c,t} = \beta \cdot \text{TransitLoss}_{c,t} + X'_{c,t}\gamma + \alpha_c + \delta_{p(c),t} + \varepsilon_{c,t}, \quad (7)$$

where $y_{c,t}$ is rebellion occurrence, α_c are county fixed effects, $\delta_{p(c),t}$ are province-by-year fixed effects, $X_{c,t}$ collects the climate, geography, and march-route controls, and standard errors are clustered at the county level. Table 7 reports estimates across the same sequence of specifications used in the baseline. Column (1) includes only county and year fixed effects; columns (2) through (6) add province-by-year fixed effects and the climate, geography, and march-route controls in turn. The coefficient is positive and highly significant throughout, declining modestly as province-by-year fixed effects and the full control set are introduced and then stabilizing. The fully controlled estimate implies that a complete loss of Canton’s export share raises the annual probability of rebellion by an amount close to the binary difference-in-differences estimate of Table 2. The stability of the coefficient across specifications indicates that the effect is driven by the collapse of Canton-bound trade rather than by an arbitrary pre-post break, since counties did not respond uniformly at the moment the treaty was signed, but increasingly as the trade on which they depended drained away.

The combined share in equation (6) pools two commodities whose export geographies differed. Tea moved south to Canton along the Anhui and Fujian feeder routes and the Gan-river trunk, whereas Jiangnan silk entered the same trunk from the lower Yangzi. Because the two commodities lost their Canton access on different timetables, decomposing the exposure measure by commodity provides a sharper test of whether the rebellion response tracks the disruption of the transit economy specifically. For each commodity $k \in \{\text{tea}, \text{silk}\}$ I define a corridor-specific flow-loss measure

$$\text{FLR}_{c,t}^k = \mathbf{1}\{k \text{ transits corridor of } c\} \times (1 - s_{\text{Canton},t}^k), \quad (8)$$

where the indicator assigns each treated county to the commodities historically carried on its segment of the route network, following Yao (1962), and $s_{\text{Canton},t}^k$ is Canton’s share of national exports of commodity k . Tea corridors comprise the Gan-river trunk together with the Wannan and Wuyi feeder routes; silk corridors comprise the trunk and the lower-Yangzi approach. I estimate equation (7) with FLR^{tea} and FLR^{silk} entered jointly, restricting the sample to 1830 through 1850, a window in which the movement in Canton’s commodity shares reflects treaty-port competition rather than the wartime disruption of trade after 1851.

Table 7: Continuous Exposure: Effect Scales with Canton Trade Collapse

	(1)	(2)	(3)	(4)	(5)	(6)
TransitLoss (continuous)	0.168*** (0.030)	0.103*** (0.024)	0.102*** (0.024)	0.102*** (0.024)	0.089*** (0.024)	0.089*** (0.024)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Province×Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	Yes	Yes	Yes
Geography control	No	No	No	Yes	Yes	Yes
March-route control	No	No	No	No	Yes	Yes
Mean of dependent variable	0.1559	0.1559	0.1559	0.1559	0.1559	0.1559
Number of counties	277	277	277	277	277	277
Observations	8587	8587	8587	8587	8587	8587
Adjusted R^2	0.194	0.348	0.348	0.349	0.350	0.350

Standard errors in parentheses

TransitLoss = Treated x (1 - s_Canton,t), Canton combined (tea+silk volume-weighted) share; silk share carried forward at its 1853 value for 1854-1860; clustered SE at county (OID).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A13 in Appendix reports the results across the same six specifications. Exposure on the tea corridors carries a large and highly significant coefficient throughout, while exposure on the silk corridors is small and statistically indistinguishable from zero in every column, and an F -test rejects equality of the two coefficients. The rebellion response to the Canton trade collapse therefore loads on the tea corridors, where the displaced labor force of porters and boatmen was concentrated. Extending the window to 1860 attenuates the tea coefficient, which remains significant without controls but weakens once the march-route control absorbs the geography of Taiping advance, consistent with the post-1851 port shares reflecting the wartime disruption of trade; these full-window results are reported in Appendix Table A14. I return to the muted silk-corridor response, and to its bearing on the distinction between the trade-access and labor-displacement interpretations, in Section 6.

Spillovers to non-treated counties. The disruption of the Canton trade network may have affected not only counties directly on the routes but also neighboring off-route counties through local market linkages and labor mobility. To test for such spillovers, I restrict the sample to off-route counties and estimate:

$$Y_{ct} = \sum_{k \neq k_0} \beta_k \left(\mathbf{1}\{\text{DistToRoute}_c \in \mathcal{B}_k\} \times \text{Post}_t \right) + \lambda_c + \delta_t + X_{ct}\Gamma + \varepsilon_{ct}, \quad (9)$$

where DistToRoute_c is the distance from county c 's centroid to the nearest Canton route segment, and $\mathcal{B}_k$ are distance bins (0–25, 25–50, 50–100, 100–200, and 200–400 km). The omitted category is the farthest bin (>400 km), so each β_k measures the post-1842 change in rebellion incidence for off-route counties at distance k relative to the most remote counties.

Figure 6 plots the estimated coefficients. Spillover effects are largest for counties within 25 km of the routes (approximately 0.10–0.12) and decline monotonically with distance, falling to near zero beyond 100 km. The attenuation pattern is consistent with local labor market effects: displaced porters and boatmen who could not find alternative employment near the old routes may have spread into adjacent counties, increasing competitive pressure on existing workers and raising local conflict risk within a restricted geographic radius. The estimated coefficients and their confidence intervals are reported in the Appendix.

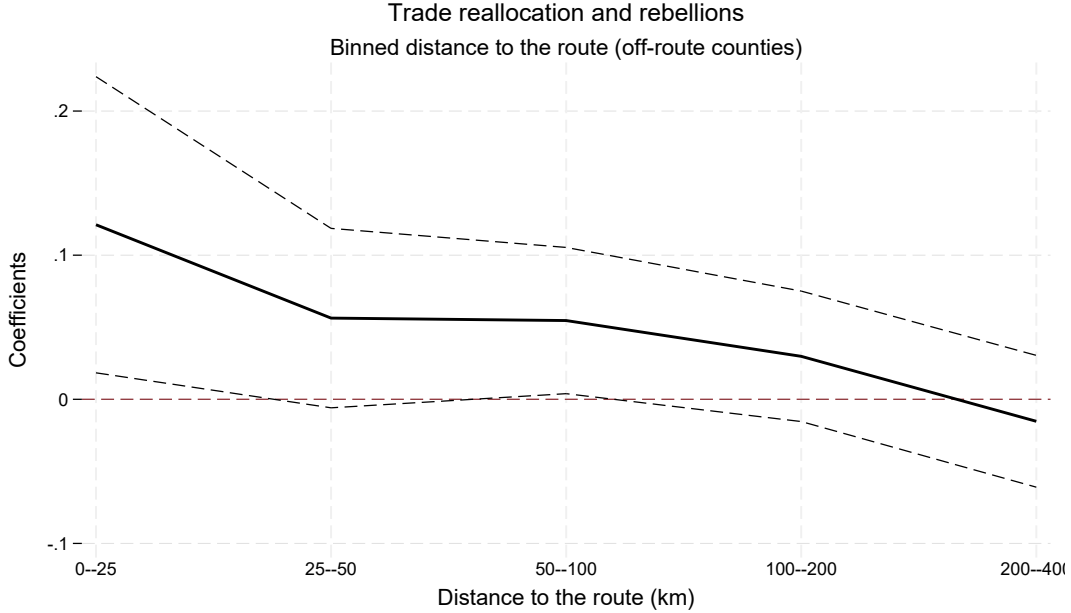


Figure 6: Spillover Effects by Distance to Historical Canton Routes

Notes: The figure plots estimated coefficients $\hat{\beta}_k$ from equation (9), with 95% confidence intervals. The sample is restricted to off-route counties. Distance bins are 0–25, 25–50, 50–100, 100–200, and 200–400 km, with > 400 km as the omitted category. Specifications include county and year fixed effects and controls for crop failure, climate anomaly, drought, and flood. Standard errors clustered at the county level.

4.4 Taiping Rebellion

The trade shock raised anti-state violence in treated counties, the form of conflict most closely tied to organized political resistance. Descriptively, this anti-state activity is concentrated in the 1850s, the decade in which the Taiping Rebellion erupted in Guangxi and expanded through the inland provinces to Nanjing. This coincidence raises the question of whether the 1842 trade shock

did not merely elevate background levels of local unrest, but specifically made treated counties more susceptible to absorption into the Taiping movement. Establishing this link would connect the disruption of the Canton system to one of the deadliest civil conflicts in recorded history, contributing to the literature on the origins of 19th-century Chinese social unrest (Kung and Ma, 2014; Bai and Jia, 2016).

I pursue this question by employing a discrete-time hazard model to examine whether trade-shock counties experienced *earlier* entry of Taiping forces, rather than merely higher average incidence. For each county, I define the event as the first year in which $\text{TaipingRebellion}_{ct} = 1$, and model the conditional probability of first entry in year t given that no entry has occurred through year $t - 1$:

$$\Pr(\text{Event}_{ct} = 1 \mid \text{Event}_{c,t-1} = 0) = 1 - \exp[-\exp(\alpha_t + \mu_{p(c)} + \beta \text{Treated}_c + X'_c\Phi + Z'_{ct}\Gamma)], \quad (10)$$

where the link function is the complementary log-log, which serves as the discrete-time analog of the continuous-time Cox proportional hazards model. The risk set opens in 1851, when the rebellion broke out. Counties exit the risk set after their first Taiping-related entry, and counties that are never affected through 1860 are right-censored. The term α_t captures the year-specific baseline hazard common to all counties, reflecting the overall intensity of Taiping spread in year t , while $\mu_{p(c)}$ are province fixed effects absorbing province-level baseline differences in susceptibility to rebellion. The vector X_c collects time-invariant geography and pre-treatment controls. These include the distance from each county centroid to Jintian Village in Guangxi (the rebellion's point of origin), terrain ruggedness, population, agricultural suitability for wheat and rice, and an indicator for any recorded conflict during the pre-shock period from 1830 to 1841. The vector Z_{ct} includes time-varying climate shocks such as flood and drought indicators with their lagged disaster counterparts, together with a spatial-diffusion control measuring the number of neighboring counties already affected by Taiping in year $t - 1$. The coefficient β measures whether trade-shock counties faced a systematically higher conditional hazard of first Taiping entry, conditional on geography, pre-existing instability, climate shocks, and the contagion of the rebellion from already-affected neighbors.

Table 8 reports the estimates across seven specifications that progressively add layers of controls. Column (1) includes only year fixed effects, providing the baseline cross-county comparison conditional on national-level hazard trends. Column (2) replaces these with the more demanding province-by-year fixed effects, which absorb any province-specific trajectory in the spread of the rebellion. Columns (3) through (7) build incrementally on the year-plus-province specification by adding geography in Column (3), pre-1842 conflict history in Column (4), climate shocks in Column (5), and two alternative spatial-diffusion measures based on queen contiguity in Column (6) and a within-50km radius in Column (7). Across all seven columns, the coefficient on Treated remains positive, statistically significant at the 1 percent level, and quantitatively stable, ranging from 0.58 to 0.69. The fact that β barely moves as these controls are layered in constitutes the cen-

tral finding of this section. The elevated hazard faced by trade-shock counties cannot be explained away by geographic proximity to the Taiping origin in Guangxi, by pre-existing local instability before the 1842 shock, by contemporaneous agrarian or climatic stress, or by the spatial contagion of Taiping forces from already-affected neighbors. Among the controls, drought enters with a large, positive, and highly significant coefficient across all columns in which it appears, consistent with the well-established link between climatic stress and rural unrest in late imperial China (Kung and Ma, 2014; Bai and Jia, 2016). The spatial-diffusion variable is also strongly significant, confirming the contagious dynamics of the rebellion as it propagated across adjacent counties.

Table 8: Trade Shock and First Taiping-Related Activity: Discrete-Time Hazard Model

	First Taiping-related activity (cloglog hazard)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated	0.6938*** (0.1725)	0.5227*** (0.1478)	0.4188*** (0.1407)	0.4259*** (0.1444)	0.4838*** (0.1431)	0.4749*** (0.1475)	0.4873*** (0.1459)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Geography controls	No	No	Yes	Yes	Yes	Yes	Yes
Pre-1842 conflict control	No	No	No	Yes	Yes	Yes	Yes
Climate controls	No	No	No	No	Yes	Yes	Yes
Spatial diffusion (Queen)	No	No	No	No	No	Yes	No
Spatial diffusion (50 km radius)	No	No	No	No	No	No	Yes
Mean of dependent variable	0.1284	0.1284	0.1284	0.1284	0.1284	0.1284	0.1284
Number of counties	277	277	277	277	277	277	277
Observations	1,589	1,589	1,589	1,589	1,589	1,589	1,589
Log-likelihood	-555.09	-485.18	-477.91	-476.71	-465.17	-460.61	-463.79

Notes: The sample consists of 277 counties in four provinces (Anhui, Jiangxi, Fujian, and Guangdong) at risk of first Taiping-related activity from 1851 onward, of which 204 experience the event and 73 are right-censored at the panel’s end in 1860. The outcome variable is an indicator equal to one in the first year a county exhibits any Taiping-related activity in the panel ($\text{TaipingRebellion}_{ct} = 1$), and zero for all prior years in the risk set; counties exit the risk set after their first event. Estimates are complementary log-log (cloglog) coefficients, interpretable as log hazard ratios; $\exp(\beta)$ gives the proportional hazard ratio. *Treated* equals one for counties on the pre-1842 inland trade routes to Canton. Column (1) includes year fixed effects; Columns (2)–(7) add province fixed effects to the year fixed effects. The geography control set comprises terrain ruggedness and the geodesic distance from each county centroid to the Taiping 1851–1853 military advance route. The pre-1842 conflict control is an indicator for any recorded conflict in 1830–1841. Climate controls comprise current-year drought and flood indicators plus lagged disaster indicators at $t - 1$ and $t - 2$. Spatial diffusion controls are the $t - 1$ lagged count of queen-contiguity neighbors (Column 6) or within-50-km-radius counties (Column 7) already exhibiting Taiping-related activity. Standard errors clustered at the county level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Together, the TWFE and hazard results establish that the 1842 trade shock did more than raise the general level of local unrest. It specifically increased counties’ vulnerability to absorption into the Taiping Rebellion. Trade-shock counties were more than twice as likely to experience Taiping-related activity after 1851, and they faced a substantially higher conditional hazard of first Taiping entry. This finding links the disruption of the Canton trade network to the origins of one of the nineteenth century’s most consequential political upheavals. It also suggests that the displacement

of transport labor in the 1840s created conditions, namely unemployed men organized into existing social networks of secret societies and local militias, that the Taiping movement exploited as it spread northward through the interior provinces.

5 Robustness Check

5.1 Alternative Estimation Techniques

The baseline specification uses a binary indicator for rebellion as the dependent variable, which treats all county-year events identically regardless of county population size. A larger county recording one rebellion incident represents a lower incidence rate than a smaller county recording the same event. To address this concern, I construct a population-normalized measure of rebellion intensity and re-estimate the baseline specification.

The construction proceeds in two steps. First, I compute a county-year incident rate by dividing the raw rebellion count by the county-level population estimate of [Zheng and Wu \(2026\)](#) for 1820:

$$r_{ct} = \frac{\text{Incidents}_{ct}}{\text{Population}_{c,1820}}. \quad (11)$$

Second, because many county-years record zero rebellions, I apply an inverse hyperbolic sine (IHS) transformation to accommodate the spike of mass at zero while approximating a logarithmic scaling for positive values:

$$Y_{ct} = \text{asinh}(\alpha \cdot r_{ct}), \quad (12)$$

where $\alpha > 0$ is a scaling constant. Because [Zheng and Wu \(2026\)](#) provide population directly at the county level, no imputation from prefecture totals is required; the estimation sample retains 274 of the 277 baseline counties (8,494 county-year observations), the three unmatched counties being excluded.

Table 9 reports the results. Across all six specifications, the coefficient on $\text{Treated} \times \text{Post-1842}$ is positive, precisely estimated, and stable. The preferred estimate in Column (2), which includes province-by-year fixed effects, is 0.0502, significant at the 1 percent level. The estimates in Columns (3)–(6), which sequentially add time-varying controls, are virtually unchanged at approximately 0.045. Two features of the results are noteworthy. First, the coefficient is somewhat smaller in magnitude than the baseline binary estimate of approximately 0.11. This compression is expected: Table 3.3 shows that treated counties are somewhat more populous than control counties on average, so the per-capita rebellion rate rises by less than the raw incidence probability. Second, the sample restriction to 274 counties does not qualitatively alter the conclusion: the effect of trade-access loss on rebellion intensity is robust to population normalization, confirming that the baseline result is not driven by county size differences between treated and control groups.

Table 9: Trade Shock and Rebellions: Population-Normalized Dependent Variable

	Rebellions (IHS normalized by 1820 population)					
	(1)	(2)	(3)	(4)	(5)	(6)
Treated $\times$ Post-1842	0.0542*** (0.0121)	0.0496*** (0.0130)	0.0494*** (0.0130)	0.0501*** (0.0128)	0.0415*** (0.0131)	0.0424*** (0.0130)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Province $\times$ Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	No	No	Yes
Geography controls	No	No	No	Yes	No	Yes
March-route control	No	No	No	No	Yes	Yes
Mean of dependent variable	0.063	0.063	0.063	0.063	0.063	0.063
Number of counties	274	274	274	274	274	274
Observations	11234	11234	11234	11234	11234	11234
Adjusted R^2	0.158	0.239	0.239	0.240	0.243	0.244

Notes: The dependent variable is the inverse hyperbolic sine transformation of the rebellion count normalized by the county-level population estimate in 1820 (equation 12). County population in 1820 is taken from the county-level estimates of [Zheng and Wu \(2026\)](#), which build on prefecture totals in [Cao \(2001\)](#) and household shares in [Liang \(1980\)](#). Three counties without a matched population estimate are excluded, yielding 274 counties and 8,494 county-year observations. Treated $\times$ Post-1842 equals one for counties lying on the pre-1842 inland routes to Canton in years ≥ 1842 . All columns include county fixed effects. Column (1) includes year fixed effects; Columns (2)–(6) include province $\times$ year fixed effects and sequentially add controls for crop failure, climate anomaly, drought, and flood. Standard errors, clustered at the county level, in parentheses.

5.2 Ruling Out Import Competition in Handicraft Textiles

One possible alternative explanation for the baseline results is increased competition from imported textile products. Following the opening of the treaty ports, machine spun yarn and machine woven cloth entered China through the new ports, and the resulting pressure on domestic spinners and weavers has long featured prominently in accounts of the late Qing economy ([Peng, 1957](#)). If counties located along the Canton routes were also concentrated in regions engaged in handicraft textile production, the estimated treatment effect might capture distress in the textile economy caused by foreign imports rather than the decline of the transit economy. This subsection evaluates that possibility directly.

I measure exposure to textile production using prefecture level indicators for the documented production of textiles and chintz, which I assign to all counties within the corresponding prefecture. This measure captures participation in a regional textile economy rather than production within

a particular county, since handicraft textile activity relied on market networks that connected spinning households, market towns, and cloth merchants across prefectural boundaries. Of the 277 counties in the negative exposure sample, 153 belong to prefectures with documented textile or chintz production. Importantly, textile exposure is almost entirely unrelated to Canton route exposure. Approximately 55 percent of both treated and control counties are located in textile producing prefectures. The geographic correlation required for textile competition to confound the baseline estimates is therefore absent.

Table 10 reports four specifications, each of which follows the fully controlled baseline specification in column 6 of Table 2. Column (1) reproduces the baseline estimate for comparison. The estimated treatment effect is 0.109 and is statistically significant at the 1 percent level. Column (2) adds an interaction between the prefecture level textile indicator and Post1842 as a competing source of post treaty distress. The treatment coefficient remains essentially unchanged at 0.110. The textile exposure coefficient is negative at 0.025 in absolute value and is statistically significant at the 5 percent level, with a (p) value of 0.049. The negative sign indicates that textile producing prefectures experienced, if anything, a relative decline rather than an increase in rebellion following the opening of the treaty ports, so this result provides no evidence that competition from imported textiles drove the baseline effect.

Column (3) offers a more direct test by limiting the sample to the 124 counties located in prefectures with no documented production of either textiles or chintz. Import competition in handicraft textiles cannot explain an increase in rebellion in areas without a documented textile economy. The treatment coefficient in this restricted sample is 0.081 and remains statistically significant at the 1 percent level, making it nearly identical to the estimate from the full sample. This restriction is deliberately conservative because it excludes every county located in a textile producing prefecture, including counties for which there is no evidence of local textile activity.

Column (4) introduces an interaction among treatment status, Post1842, and textile exposure. If the baseline relationship were driven by distress in the textile sector, the post treaty treatment effect should be larger in textile producing prefectures. The estimated triple interaction is negative at 0.008 in absolute value and is statistically insignificant, with a (p) value of 0.83. The estimated treatment effect therefore does not differ meaningfully between textile and nontextile prefectures.

The timing of foreign textile penetration provides an additional reason to reject this interpretation. Historians generally date the substantial displacement of Chinese handicraft cloth by imported textiles to the 1860s and later, after the end of the sample period. Before 1860, imported textiles remained concentrated mainly in areas close to the treaty ports (Peng, 1957). Taken together, the lack of geographic correlation between Canton route exposure and textile production, the stability of the treatment coefficient after controlling for textile exposure, the persistence of the effect in places without documented textile production, the insignificant triple interaction, and the timing of import penetration all indicate that competition from foreign textiles does not account for the baseline findings.

Table 10: Robustness to Import Competition in Handicraft Textiles

	(1)	(2)	(3)	(4)
Treated $\times$ Post1842	0.109*** (0.018)	0.110*** (0.018)	0.119*** (0.032)	0.114*** (0.029)
Treated $\times$ Post1842 $\times$ Textile pref.				-0.008 (0.039)
Textile pref. $\times$ Post1842		-0.025** (0.013)		-0.024 (0.015)
Climate controls	Yes	Yes	Yes	Yes
Geography $\times$ Post1842	Yes	Yes	Yes	Yes
March-route $\times$ Post1851	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Province $\times$ Year FE	Yes	Yes	Yes	Yes
Observations	11356	11356	5084	11356
Within-R ²	0.0095	0.0100	0.0098	0.0100

Notes: Standard errors clustered by county in parentheses. Specification identical to Table 2 Column 6 (dependent variable: an indicator equal to one if at least one rebellion occurred in county c in year t). Textile pref. indicates the county lies in a prefecture with documented textile or chintz production. Column 3 restricts to counties in prefectures with no textile or chintz production. All columns include county fixed effects, province $\times$ year fixed effects, climate controls (drought, flood), the geography control (ruggedness $\times$ Post-1842), and the march-route control (distance to the Taiping march route $\times$ Post-1851). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Placebo Tests

I probe the validity of the research design with a placebo test on the geography of the treated routes. The concern is that the estimated effect might reflect a general response of counties along historical transport corridors to the 1842 shock, rather than something specific to the Canton trade routes. To address this, I assign a placebo treatment to counties lying on the Qing courier road network (*yilu*), digitized from the China Historical GIS (2016) Ming courier routes. The courier roads formed the empire’s official communication and transport network and were not defined by the Canton export trade, so they provide a natural placebo corridor. If the post-1842 rebellion increase were a generic feature of transport-route counties, I would expect a positive and significant coefficient on $Courier \times Post-1842$. The opposite, a null result, supports the interpretation that the effect is specific to the disruption of the Canton trade network.

I estimate the baseline DID specification, replacing the Canton-route treatment with the courier-road indicator. Table 11 reports the results. Panel A uses the full sample of 277 counties. Because the courier network overlaps geographically with the Canton routes, with roughly 55 percent of Canton-route counties also lying on courier roads, Panel B reports a cleaner placebo that excludes all Canton-treated counties, comparing courier-only counties with counties on neither network. Across all specifications in both panels, the coefficient on $Courier \times Post-1842$ is small and statistically insignificant. The absence of any placebo effect, even though the courier network is geographically entangled with the Canton routes, indicates that the rise in rebellion after 1842 was concentrated specifically in counties that lost access to the Canton trade network rather than in transport-corridor counties generally.

5.4 PSM-DID

Table 3.3 shows that treated and control counties differ in several characteristics measured before the trade shock. Counties located along the Canton routes had larger populations, experienced floods more frequently, and had somewhat higher densities of market towns. County fixed effects account for all persistent differences between counties. A remaining concern, nonetheless, is that treated counties may differ systematically from control counties in observable characteristics that are associated with subsequent changes in conflict. I address this concern by constructing a matched sample in which treated and control counties are more comparable, and then estimating the baseline specification on this sample.

I estimate the probability of exposure to the Canton routes using a cross sectional logit model with nine pre-treatment covariates: terrain ruggedness, county area, population in 1820, market town density, suitability for rice and wheat cultivation, the frequency of floods and droughts between 1830 and 1841, and the frequency of rebellion over the same period. Each treated county is then matched to the control county with the closest propensity score. I use one to one nearest neighbour matching, impose a caliper of 0.05, and do not allow controls to be reused. Of the 51 treated

Table 11: Placebo: Courier Roads and Rebellions (P1 full; P2 excl. Canton-treated)

	Rebellion occurrence (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: P1 – full sample</i>						
Courier road (placebo) $\times$ Post-1842	-0.0146 (0.0155)	0.0112 (0.0143)	0.0112 (0.0143)	0.0162 (0.0146)	0.0139 (0.0137)	0.0181 (0.0140)
Mean of dep. var.	0.1549	0.1549	0.1549	0.1549	0.1549	0.1549
Number of counties	277	277	277	277	277	277
Observations	8,586	8,586	8,586	8,586	8,586	8,586
<i>Panel B: P2 – clean placebo (excl. Canton-treated counties)</i>						
Courier road (placebo) $\times$ Post-1842	-0.0129 (0.0165)	0.0051 (0.0155)	0.0048 (0.0155)	0.0096 (0.0159)	0.0083 (0.0149)	0.0116 (0.0153)
Mean of dep. var.	0.1409	0.1409	0.1409	0.1409	0.1409	0.1409
Number of counties	226	226	226	226	226	226
Observations	7,005	7,005	7,005	7,005	7,005	7,005
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Prov. $\times$ Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	No	No	Yes
Geography controls	No	No	No	Yes	No	Yes
March-route control	No	No	No	No	Yes	Yes

Notes: Placebo test. The treatment is *Courier road* $\times$ *Post-1842*, an indicator for counties on the CHGIS v6 (2016) Ming courier-road network (), interacted with the post-1842 dummy. The dependent variable is an indicator equal to one if at least one rebellion occurred in county c in year t . Specifications mirror the baseline Canton DID columns (1)–(6). Panel A uses all counties; Panel B excludes Canton-treated counties (the clean placebo). *Post-1842* = 1 if year $\geq$ 1842 (matching the baseline definition); occurrence is set to zero during the Opium War (*OpiumWar*==1), as in the baseline. Standard errors clustered at the county level (OID) are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

counties, 42 are successfully matched to 42 distinct control counties, producing a matched sample of 84 counties. The remaining nine treated counties either lie outside the common support or have no eligible control county within the specified caliper.

Appendix Table A8 evaluates covariate balance before and after matching. Prior to matching, several characteristics display substantial standardized differences, particularly population in 1820 and the historical frequencies of floods and droughts, consistent with the imbalances reported in Table 3.3. Matching substantially improves comparability between the two groups. The mean absolute standardized bias falls by roughly three quarters, and the pseudo R^2 from the treatment model declines sharply. The likelihood ratio test strongly rejects the joint orthogonality of the covariates to treatment before matching, whereas after matching the same hypothesis cannot be rejected. None of the individual covariate differences is statistically significant in the matched sample.

The frequency of rebellion before the trade shock is already similar across treated and control counties before matching and remains balanced afterward, with a small standardized difference in absolute value. This balance is reassuring because it indicates that treated and matched control counties had similar levels of conflict before 1842. Together with the event study evidence of parallel trends, it reduces the concern that the estimated effect reflects pre-existing differences in conflict trajectories.

Appendix Table A9 estimates the baseline difference in differences specification on the matched sample. The coefficient on $Treated \times Post1842$ remains positive and statistically significant at the 1 percent level, and is very close to the full sample estimate. The effect therefore retains essentially the same magnitude after matching, despite the substantial reduction in sample size and the associated increase in the standard error. The matched sample includes both treated and control counties in each of the four provinces, so the province by year fixed effects remain identified. The stability of the coefficient across the full and matched samples indicates that observable differences between counties on and off the Canton routes do not account for the baseline result.

5.5 Changes-in-Changes

The baseline difference in differences estimator imposes a homogeneous treatment effect across all observations. One concern with this assumption is that counties lying on the Canton trade corridor may differ systematically from off corridor counties in ways that interact with the post 1842 environment, so that a common national shock could provoke a differential response unrelated to the corridor mechanism itself. The baseline specification already mitigates this concern within the difference in differences framework, both by absorbing province by year effects and by allowing flexible time effects of pre treatment rebelliousness, and the event study reported above confirms that treated and control counties followed parallel trends before 1842. This section provides further reassurance by employing two alternative procedures that account for systematic differences between the treated and control groups in ways the standard estimator does not.

The first procedure is the changes in changes (CIC) estimator of [Athey and Imbens \(2006\)](#), which generalizes the standard difference in differences model. Rather than comparing mean outcomes across the two groups, the CIC estimator recovers the entire distribution of the counterfactual outcome and estimates quantile treatment effects nonparametrically, comparing this counterfactual distribution to the realized post period distribution for the treated group. The method imposes no functional form on the outcome distribution and permits the treated and control groups to differ in the distribution of unobservable characteristics in arbitrary ways, provided that distribution is time invariant within each group. A positive effect that survives this relaxation is therefore informative regardless of how the two groups differ at baseline.

I follow the two step procedure of [Athey and Imbens \(2006\)](#). The first step uses ordinary least squares to partial out the covariates and the high dimensional fixed effects, namely county fixed effects, year fixed effects, and province by year fixed effects, together with the baseline controls. The second step applies the unconditional CIC estimator to the residuals to recover the quantile treatment effects, with standard errors from 100 bootstrap replications. The estimates are reported in Appendix Table [A10](#).

Two patterns emerge. First, the corridor shock raises rebellion intensity across the interpretable range of the distribution. The overall average treatment effect is positive and significant, and the quantile treatment effects are positive and significant at every quantile from the 25th upward. This confirms the positive effect found in the baseline and shows that it does not depend on comparing group means. I caution that the rebellion intensity outcome has a large mass at zero, since most county years record no incident, so the quantile treatment effects below the 25th percentile fall within this zero mass region, where the estimator carries little information. I report them in Appendix Table [A10](#) for completeness but do not interpret them, and I restrict attention to the 25th percentile and above.

Second, the effect is largest in the upper tail of the distribution, well above the values near the median. The corridor shock therefore did not raise rebellion uniformly. Instead it shifted the upper portion of the intensity distribution, pushing the most affected counties toward substantially higher levels of unrest. This concentration in the upper tail is consistent with the displacement mechanism, since the collapse of corridor traffic released transport labor in concentrated locations along the route, and where local conditions were already combustible this surplus translated into more intense and larger scale uprisings rather than a uniform increase everywhere.

5.6 Synthetic Difference-in-Differences

The second procedure constructs a synthetic counterfactual for the treated group. Whereas the CIC estimator allows the distribution of unobservables to differ across groups in arbitrary ways, the synthetic approach addresses comparability by building a weighted combination of control counties that tracks the evolution of the treated group’s outcome before 1842. The post period trajectory of this synthetic control then serves as a projection of what the corridor counties would have

experienced absent the trade shock.

The classical synthetic control method of [Abadie, Diamond, and Hainmueller \(2010\)](#) was designed for settings with a long pre treatment series, which my panel does not provide, since the corridor shock occurs in 1842 and the pre period spans only the years 1830 to 1841. I therefore employ the synthetic difference in differences estimator, which combines unit and time weighting with a difference in differences structure and is more robust to short pre treatment panels than the classical estimator. Treatment is block assigned in 1842 to all corridor counties, and I compute placebo, bootstrap, and jackknife standard errors. The results appear in Appendix Table [A12](#).

The estimated average treatment effect on the treated is positive and significant at the one percent level using all control counties as donors, and it is stable across the three inference procedures. The point estimate and significance match the baseline difference in differences results in both sign and magnitude. To guard against contamination from control counties that lie close to the corridor and may absorb spillovers, I also estimate a specification that drops the control counties within 52 kilometers of the route, a threshold just above the maximum distance among treated counties. The effect is essentially unchanged and remains significant at the one percent level, and the slight increase is consistent with the removal of contaminated comparison units.

Appendix Figure [A1](#) plots the trajectories of the treated and synthetic control groups. The two series diverge after 1842, with corridor counties lying systematically above their synthetic counterfactual throughout the post period. Two features of the figure deserve comment. First, rebellion rises in the synthetic control group as well during the 1850s, reflecting the general political instability of the mid nineteenth century, most notably the Taiping upheaval. Corridor counties, however, begin to separate from their synthetic counterfactual earlier, around 1850 to 1851, and remain above it thereafter. Were the corridor counties simply more sensitive to nationwide turmoil, the two series would have risen in step; the earlier and larger increase among treated counties instead points to a corridor specific shock. Second, I acknowledge a limitation. Because the pre treatment window is short and the outcome is sparse, the synthetic control’s pre period fit is imperfect, with a pre 1842 prediction error that is non-negligible relative to the estimated treatment effect. I therefore treat the synthetic difference in differences estimate as corroborating evidence rather than as primary inference, and the two way fixed effects difference in differences specification remains my main result.

Taken together, the two procedures generalize the standard difference in differences methodology by allowing for systematic differences between corridor and off corridor counties, and both confirm that the positive effect of the trade shock on rebellion is not an artifact of comparing dissimilar groups.

6 Discussion of Mechanisms

The results in Section 4 establish that counties exposed to the collapse of the Canton trade network experienced a substantial increase in rebellion probability after 1842. This section discusses the economic channels through which this effect may have operated. Two candidate mechanisms are considered: an opportunity cost channel operating through the displacement of transport labor; and a state capacity channel operating through the erosion of fiscal and administrative resources. I draw where possible on the regression evidence presented in Section 4 to assess the relative plausibility of each channel, and I outline the empirical work required to test them more formally.

6.1 Opportunity Cost Channel

The opportunity cost channel is the most natural explanation for the empirical results. Under this interpretation, the collapse of the Canton trade network destroyed the livelihoods of the large population of transport workers who had sustained themselves through the route economy, including porters who carried goods over mountain passes and boatmen who navigated the river systems linking the interior to Guangzhou. With their primary source of income eliminated and few alternative opportunities in a largely pre-industrial rural economy, these workers faced a sharp decline in the cost of participating in collective violence. As [Wakeman \(1966\)](#) documents, displaced transport workers became a ready pool of recruits for secret societies and armed resistance in the years following the treaty. The prediction of the opportunity cost channel is that the treatment effect should be larger where the trade-dependent labor force was larger, that is, in counties more deeply embedded in the transport economy of the Canton system ([Becker, 1968](#)).

I do not have direct evidence on this channel in the form of employment records or wage data for porters and boatmen at the county-year level; such data are not available for mid-nineteenth-century China. The empirical evidence presented in Section 4 is therefore best interpreted as circumstantial but coherent support for the opportunity cost interpretation, rather than a formal test.

Several pieces of evidence point in this direction. First, the null result for positively exposed counties argues against an alternative rapacity interpretation, under which greater trade activity generates contestable rents and stimulates conflict among rival claimants ([Grossman, 1991](#)). If rent-seeking were the operative mechanism, counties gaining access to the new treaty port trade should also experience increased conflict, a prediction that the data clearly reject. The asymmetry between negative and positive exposure instead points toward a mechanism rooted in income loss rather than rent creation.

Second, the heterogeneity results in Table 5 show that the treatment effect is concentrated among transport-exposed counties, whose primary function in the Canton network was to facilitate the movement of goods, relative to production-exposed counties whose economies centered on commodity cultivation. The difference is marginally significant, with a p -value of 0.10 in the preferred

specification, and its direction is consistent with the opportunity cost channel. Transport workers faced a near-complete destruction of demand for their services following the reallocation of trade toward coastal shipping, whereas commodity producers could in principle reallocate land and labor to other markets over time.

Third, the economic intensity results in Table 6 bear on the same logic, though in a way that sharpens rather than simply echoes it. As reported in Section 4.3, the treatment effect does not scale with pre-1842 commercial density: the tercile profile shows no monotone pattern, and equality of the first and third terciles cannot be rejected for either intensity measure, so exposure is well described by corridor location itself rather than by the breadth of local commerce within treated counties. This pattern is consistent with the transit labor channel, in which the displaced livelihoods of porters, boatmen, and innkeepers were tied to the through traffic of the corridor rather than to the density of commercial activity, so that displacement followed corridor placement rather than local market size. It contrasts with [Cao and Chen \(2022\)](#), who find that the rebellion effect of the Grand Canal’s closure is significantly stronger in counties with a higher share of market towns within ten kilometers of the canal. The difference is instructive: the Grand Canal carried domestic tribute grain whose handling concentrated in commercial towns, whereas the Canton route economy rested on a transport workforce strung along mountain passes and river segments, for whom proximity to the corridor, not local commercial density, determined exposure to the shock.

Fourth, the geographic attenuation of spillover effects documented in Figure 6, which decays from approximately 0.10 to 0.12 within 25 km of the routes to near zero beyond 100 km, is consistent with local labor market spillovers. Displaced workers who could not find alternative employment near the route corridor may have migrated into adjacent counties, generating downward pressure on local wages and heightened social strain within a restricted geographic radius. The localized nature of the spillovers also argues against explanations based on broader provincial fiscal or political shocks, which would be expected to generate more diffuse geographic patterns.

Fifth, I provide more direct evidence on the link between trade disruption and secret society activity, the recruitment channel emphasized by [Wakeman \(1966\)](#). [Qin and Meng \(2008\)](#) describes this process for Guangdong after the opening of the treaty ports: once Canton lost its monopoly over foreign trade, the small merchants and the laborers who had made their living along the trade routes were thrown out of work in large numbers, and many of these destitute people turned to the Tiandihui (Heaven and Earth Society) for support, producing a rapid expansion of secret society organization in the province. This account maps directly onto the opportunity cost mechanism: boatmen and porters who lost their livelihoods along the Canton routes became recruits for secret societies. Motivated by this account, I test whether secret society activity after 1842 was systematically more prevalent in the treated corridor counties.

I assemble county-level records of post-1842 Tiandihui foundings from [Qin and Meng \(2008\)](#)

and [Qin \(2005\)](#) and estimate a cross-sectional Poisson regression of the form

$$\log E[\text{Foundings}_c] = \beta \text{Treated}_c + X'_c \gamma + \delta_{p(c)}, \quad (13)$$

where Foundings_c is the number of secret society foundings recorded in county c after 1842, Treated_c indicates a county on the pre-1842 Canton routes, X_c controls for log population, market town density, and terrain ruggedness, and $\delta_{p(c)}$ is a province fixed effect. The exponentiated coefficient $\exp(\beta)$ is the incidence-rate ratio, the proportional difference in the expected number of foundings between treated and control counties. [Table 12](#) reports the results. In the preferred specification, a treated county had roughly 2.8 times the expected number of post-1842 secret society foundings relative to a control county in the same province with comparable population, commercial density, and terrain ($p < 0.01$), and the positive association is stable across Poisson and ordinary least squares estimators and across the narrower 1842 to 1860 window. Foundings are most numerous in Guangzhou prefecture, the terminus of the Canton corridor and the jurisdiction most exposed to the post-1842 reallocation of trade, exactly as the mechanism would predict. This is a cross-sectional association rather than a causal estimate, since province fixed effects cannot fully separate the corridor from the broader geography of secret society activity, and I therefore read it as suggestive corroboration rather than a formal test. Taken together with the qualitative account in ? and the indirect evidence above, it lends further weight to the interpretation that the trade shock raised conflict by displacing trade-dependent labor.

6.2 State Capacity

A further possibility is that the trade shock operated through a weakening of state capacity. The Canton trade supported private livelihoods, but it also supplied the Qing state with considerable fiscal revenue. Customs duties levied on goods moving through Guangzhou and along the inland routes fed into provincial finances and the imperial household treasury. As overseas commerce shifted toward the new treaty ports after 1842, the revenue tied to the Canton trade may have fallen. Diminished fiscal resources could in turn have eroded the government's capacity to maintain order, by constraining the funds available for military garrisons, yamen staff, and the granaries that underpinned famine relief and social insurance ([Cao and Chen, 2022](#)).

Two features of the setting make this channel unlikely to drive the results. A large portion of Canton customs revenue flowed to the imperial household rather than staying with local governments, so the decline of the Canton trade did not map directly onto an equivalent loss of county level fiscal resources. In addition, the preferred specifications absorb province by year fixed effects, which net out fiscal and administrative shocks common to a province in a given year, so that identification rests on comparisons among counties within the same province and year. For state capacity to account for the baseline estimates, administrative capacity would have had to weaken more steeply in counties along the former Canton routes than in their within province neighbors,

Table 12: Trade Shock and Post-1842 Secret-Society Founding: County Cross-Section

	(1) Poisson	(2) Poisson	(3) OLS	(4) Poisson	(5) Poisson
Treated (Canton route)	1.0287*** (0.3829)	1.1921*** (0.3612)	0.1850** (0.0915)	1.0605** (0.4356)	1.0769* (0.5546)
Incidence-rate ratio $\exp(\beta)$	2.798	3.294		2.888	2.936
Province FE	Yes	Yes	Yes	No	Yes
Prefecture FE	No	No	No	Yes	No
Controls	Yes	No	Yes	Yes	Yes
Sample window	Post-1842	Post-1842	Post-1842	Post-1842	1842-1860
Observations	274	277	274	127	215
Pseudo R^2 / R^2	0.121	0.102	0.081	0.137	0.122

Dependent variable: the number of post-1842 Tiandihui (secret-society) foundings per county (columns 1–4); column 5 counts foundings during 1842–1860. Sample = the 277 baseline counties (Anhui, Jiangxi, Guangdong, Fujian); counties with no founding take a true zero. Columns 1, 2, 4 and 5 are `ppmlhdfc` HDFE-Poisson estimates with HC1-robust standard errors; column 3 is OLS with the same robust SE. The incidence-rate ratio is $\exp(\beta)$, the multiplicative effect on the expected count: the headline column 1 implies a treated (Canton-route) county had roughly 2.8× the expected number of foundings. Controls = log county population, market-town density, and ruggedness. Column 4 absorbs prefecture fixed effects; `ppmlhdfc` drops all-zero prefecture groups (separation), so its sample is reduced (within-prefecture treatment variation exists in only about 12 prefectures) and the estimate should be read as suggestive. Foundings are most numerous in Guangzhou prefecture, the corridor terminus most exposed to the post-1842 reallocation of the Canton trade, consistent with the proposed mechanism. Estimates are cross-sectional associations, not causal.

Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and to do so specifically after 1842. That is a stringent requirement.

I nonetheless test the mechanism directly. If state capacity buffered the trade shock, counties with stronger pre-1842 administration should have been better positioned to suppress emerging unrest, and the treatment effect should be smaller where initial capacity was greater. I capture baseline administrative capacity with three proxies.

The first is the county’s regional importance rating under the Qing “chong fan pi nan” classification, which graded jurisdictions from one to four according to their administrative weight and the difficulty of governing them. More highly rated posts were treated as more demanding and tended to receive abler officials and larger administrative resources (Liu and Zhang, 2025; Zhang, 2011). The ratings were fixed through three empire wide revisions completed between 1733 and 1747 and changed little afterward, so they were set well before the trade shock. I take the rating from personnel records spanning 1760 to 1798 and limit the analysis to the 260 sample counties for which it is available.

The second proxy indicates whether a county served as a prefecture capital. Such counties housed higher tiers of government and typically held denser concentrations of bureaucratic and military resources. The third proxy is the presence of Green Standard troops before 1842, a more direct gauge of the coercive resources at local authorities’ disposal. I take battalion troop quotas from Luo (1984) and assign them to counties using the headquarters assignment procedure of Cao and Chen (2022). I build two versions: a headquarters measure that gives the full battalion quota to the county holding its headquarters, and a shared measure that splits the quota across all counties in the battalion’s jurisdiction. Counties with no recorded garrison take a value of zero, so the military measures span all 277 counties in the baseline sample.

I interact each proxy with treatment status and the post 1842 indicator. For the regional importance rating, I estimate

$$Y_{ct} = \beta_1 (\text{Treated}_c \times \text{Post}_t) + \beta_2 (\text{Treated}_c \times \text{Post}_t \times \text{Rank}_c) + \beta_3 (\text{Post}_t \times \text{Rank}_c) + \lambda_c + \varphi_{p(c),t} + X_{ct}\Gamma + \varepsilon_{ct}, \quad (14)$$

where Rank_c is the county’s regional importance rating, centered on its sample mean. The prefecture capital specification replaces Rank_c with the indicator PrefCap_c . For military capacity, I estimate

$$Y_{ct} = \beta_1 (\text{Treated}_c \times \text{Post}_t) + \beta_2 (\text{Treated}_c \times \text{Post}_t \times \text{Soldiers}_c) + \beta_3 (\text{Post}_t \times \text{Soldiers}_c) + \lambda_c + \varphi_{p(c),t} + X_{ct}\Gamma + \varepsilon_{ct}, \quad (15)$$

where $\text{Soldiers}_c = \text{asinh}(\text{troops}_c/100)$ is the inverse hyperbolic sine of the county troop quota, centered on its sample mean. I estimate this specification with both the headquarters and shared measures. The capacity variables and their interactions with Treated_c are time invariant and are absorbed by the county fixed effects. The coefficient of interest is β_2 , which the state capacity

mechanism predicts should be negative, since stronger initial capacity should dampen the effect of the trade shock on rebellion.

Table 13 presents the results. The main difference in differences coefficient stays positive and highly significant throughout, and for a county at the average value of each capacity measure the estimated effect remains substantial. None of the interaction estimates lends convincing support to the state capacity mechanism.

For the regional importance rating, the interaction is positive and significant only at the ten percent level. Greater administrative importance is associated with a larger rather than a smaller increase in rebellion, contrary to what the state capacity mechanism predicts. A more plausible reading is that administratively important counties often sat on active commercial corridors and held larger concentrations of workers whose livelihoods depended on trade, so that the positive interaction is consistent with the transport exposure gradient documented in Section 4.3.

The interaction with prefecture capital status is negative in sign, which would be consistent with a cushioning effect, but it is statistically insignificant and therefore offers no reliable evidence that prefecture capitals contained the rise in rebellion more effectively. The Green Standard troop measures are similarly weak: under both the headquarters and the shared assignment, the interaction is statistically indistinguishable from zero, so the size of the local Green Standard establishment does not appear to moderate the effect of the trade shock. Taken together, the positive interaction for the regional importance rating and the null results for prefecture capital status and military presence provide little support for declining state capacity as the principal mechanism.

A more direct test would require county or prefecture level data on fiscal revenue, military strength, or administrative staffing before and after 1842. The Green Standard quotas used here describe the military establishment before the shock but do not vary over time, and richer annual records are only now emerging through the digitization of Qing archival materials. I therefore leave a direct study of changes in local fiscal and administrative capacity to future research. Even so, the evidence presented here, together with the absence of any increase in conflict among positively exposed counties documented in Section 4.2, points away from an erosion of state capacity and toward the displacement of workers whose livelihoods depended on the Canton trade network.

7 Conclusion

This paper has studied the causal effect of trade disruption on local social conflict using the 1842 Treaty of Nanjing as a source of exogenous variation in counties' exposure to the collapse of the Canton trade network. The main finding is that counties on the pre-1842 inland routes to Canton experienced a significant and persistent increase in rebellion probability after 1842, relative to other counties in the same provinces. The effect is robust to alternative specifications and dependent variable constructions. Counties positively exposed to the new treaty ports show no comparable increase; these patterns are jointly consistent with an opportunity-cost mechanism

Table 13: Trade Shock and Rebellion: Heterogeneity by Baseline Administrative Capacity

	(1)	(2)	(3)	(4)
Treated $\times$ Post1842	0.0751*** (0.0186)	0.0962*** (0.0224)	0.0860*** (0.0189)	0.0887*** (0.0193)
Treated $\times$ Post $\times$ Quefen	0.0411* (0.0214)			
Treated $\times$ Post $\times$ Prefecture capital		-0.0373 (0.0399)		
Treated $\times$ Post $\times$ Soldiers (asinh, HQ)			0.0101 (0.0115)	
Treated $\times$ Post $\times$ Soldiers (asinh, share)				0.0019 (0.0109)
County FE	Yes	Yes	Yes	Yes
Province $\times$ Year FE	Yes	Yes	Yes	Yes
Six kitchen-sink controls	Yes	Yes	Yes	Yes
Observations	8,060	8,060	8,587	8,587
Counties (clusters)	260	260	277	277
Adjusted R^2	0.352	0.352	0.351	0.351

Dependent variable: `occurrence`, a binary indicator for any rebellion event in the county-year. Columns (1)–(2) restrict the sample to the 260 counties with a non-missing CGED-Q quefen rank (17 counties lacking a quefen record are excluded); columns (3)–(4) use all 277 counties, because the military capacity proxy has no missing values (counties with no Green Standard garrison are true zeros). Panel 1830–1860, balanced. Standard errors clustered by county in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1) centers `quefen_y1798_num` at its in-sample mean (1.91), so `Treated  $\times$  Post1842` is the DID at the average county. The military capacity terms in columns (3)–(4), `asinh(soldier/100)`, are likewise centered at their in-sample means (HQ: 0.97; share: 0.89). `soldier_hq` assigns each Green Standard battalion’s troop quota to its single garrison (HQ) county; `soldier_share` instead splits each battalion’s quota equally across all counties it administers. Troop data are from Luo Ergang, *Lüying bing zhi*, with counties assigned following the rule of Cao & Chen (2022). All specifications include county FE, province $\times$ year FE, and controls (`cropfailure`, `climateanomaly`, `flood.broad`, `drought.broad`, `ruggedness  $\times$  post1842`, `dist.march.route_km  $\times$  post1851`).

driven by labor displacement rather than a rapacity mechanism driven by contestable rents.

The effect holds when conflict is disaggregated by motive, appearing for both anti-social and anti-state violence, which indicates that the shock raised disorder broadly rather than along a single dimension. The anti-state response, descriptively concentrated in the 1850s, coincides with the outbreak of the Taiping Rebellion. Trade-shock counties were substantially more likely to experience Taiping-related conflict and faced a higher conditional hazard of first Taiping entry, establishing a link between the disruption of international trade and the origins of one of the most consequential civil conflicts in recorded history.

More broadly, the results illustrate how international trade reallocation, even when it expands aggregate trade volumes, can have deeply destabilizing consequences for the specific communities and workers embedded in old networks. Understanding these distributional effects of trade integration remains important for contemporary debates about the social costs of de-globalization and the political economy of trade adjustment.

References

- Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. “Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California’s Tobacco Control Program.” *Journal of the American Statistical Association* 105 (490): 493–505.
- Athey, Susan, and Guido W. Imbens. 2006. “Identification and Inference in Nonlinear Difference-in-Differences Models.” *Econometrica* 74 (2): 431–97.
- Bai, Ying, and James Kai-sing Kung. 2011. “Climate Shocks and Sino-Nomadic Conflict.” *Review of Economics and Statistics* 93 (3): 970–81.
- Bai, Ying, and Ruixue Jia. 2016. “Elite Recruitment and Political Stability: The Impact of the Abolition of China’s Civil Service Exam.” *Econometrica* 84 (2): 677–733.
- Bazzi, Samuel, and Christopher Blattman. 2014. “Economic Shocks and Conflict: Evidence from Commodity Prices.” *American Economic Journal: Macroeconomics* 6 (4): 1–38.
- Becker, Gary S. 1968. “Crime and Punishment: An Economic Approach.” *Journal of Political Economy* 76 (2): 169–217.
- Cao, Shuji. 2001. *The Population History of China, Vol. 5*. Shanghai: Fudan University Press. [In Chinese].
- Cao, Yiming, and Shuo Chen. 2022. “Rebel on the Canal: Disrupted Trade Access and Social Conflict in China, 1650–1911.” *American Economic Review* 112 (5): 1555–1590.

- Chinese Academy of Meteorological Sciences. 1981. *The Atlas of the Distribution of Droughts and Floods in China over the Past 500 Years*. Beijing: SinoMaps Press. [In Chinese].
- Dell, Melissa, Benjamin Feigenberg, and Kensuke Teshima. 2019. “The Violent Consequences of Trade-Induced Worker Displacement in Mexico.” *American Economic Review: Insights* 1 (1): 43–58.
- Donaldson, Dave, and Richard Hornbeck. 2016. “Railroads and American Economic Growth: A ‘Market Access’ Approach.” *Quarterly Journal of Economics* 131 (2): 799–858.
- Dube, Oeindrila, and Juan F. Vargas. 2013. “Commodity Price Shocks and Civil Conflict: Evidence from Colombia.” *Review of Economic Studies* 80 (4): 1384–1421.
- Earth Resources Observation and Science Center/US Geological Survey/US Department of the Interior. 1997. “USGS 30 ARC-second Global Elevation Data, GTOPO30.” Research Data Archive at the National Center for Atmospheric Research, Computational and Information Systems Laboratory, Boulder, CO. <https://doi.org/10.5065/A1Z4-EE71>.
- Fortune, Robert. 1999. *Two Visits to the Tea Countries of China and the British Tea Plantations in the Himalaya*. Elibron Classics Series. Adegri Graphics LLC. (Original work published 1853).
- Grossman, Herschell I. 1991. “A General Equilibrium Model of Insurrections.” *American Economic Review* 81 (4): 912–21.
- Hao, Zhixin, Mengxin Bai, Jingyun Zheng, and Quansheng Ge. 2023. “A Hydroclimate Gridded Dataset for Eastern China over the Last Millennium.” *Scientific Data* 10 (1): 258.
- Harvard Yenching Institute, and Fudan Center for Historical Geography. 2007. “CHGIS, Version 4.” Harvard University and Fudan University. <https://sites.fas.harvard.edu/~chgis/data/chgis/v4/>.
- International Institute for Applied Systems Analysis, and Food and Agriculture Organization of the United Nations (IIASA and FAO). 2012. “Global Agro-Ecological Zones (GAEZ v3.0).” Food and Agriculture Organization of the United Nations. <https://gaez.fao.org/Main.html>.
- Jia, Ruixue. 2014. “The Legacies of Forced Freedom: China’s Treaty Ports.” *The Review of Economics and Statistics* 96 (4): 596–608.
- Kung, James Kai-sing, and Chicheng Ma. 2014. “Can Cultural Norms Reduce Conflicts? Confucianism and Peasant Rebellions in Qing China.” *Journal of Development Economics* 111: 132–49.
- Liang, Fangzhong. 1980. *Historical Statistics on Hukou, Land and Land Tax of China*. Beijing: Zhonghua Shuju. [In Chinese].

- Liu, Kevin Zhengcheng, and Xiaoming Zhang. 2025. “Discretion, Talent Allocation, and Governance Performance: Evidence from China’s Imperial Bureaucracy.” *Journal of Development Economics* 172: 103391.
- Luo, Ergang. 1984. *Book of the Green Standard Army* [UTF8gbns]. Beijing: Zhonghua Shuju. [In Chinese].
- Mann, Michael E., Zhihua Zhang, Scott Rutherford, Raymond S. Bradley, Malcolm K. Hughes, Drew Shindell, Caspar Ammann, Greg Faluvegi, and Fenbiao Ni. 2009a. “Global Signatures and Dynamical Origins of the Little Ice Age and Medieval Climate Anomaly.” *Science* 326 (5957): 1256–60.
- Mann, Michael E., Zhihua Zhang, Scott Rutherford, Raymond S. Bradley, Malcolm K. Hughes, Drew Shindell, Caspar Ammann, Greg Faluvegi, and Fenbiao Ni. 2009b. “Supplementary Material for: Global Signatures and Dynamical Origins of the Little Ice Age and Medieval Climate Anomaly.” American Association for the Advancement of Science. https://www.science.org/doi/suppl/10.1126/science.1177303/suppl_file/multiproxyspatial09.zip.
- Miguel, Edward, Shanker Satyanath, and Ernest Sergenti. 2004. “Economic Shocks and Civil Conflict: An Instrumental Variables Approach.” *Journal of Political Economy* 112 (4): 725–53.
- Nunn, Nathan, and Diego Puga. 2012. “Ruggedness: The Blessing of Bad Geography in Africa.” *Review of Economics and Statistics* 94 (1): 20–36.
- Peng, Zeyi. 1957. *Source Materials on the History of Handicraft Industry in Modern China (1840–1949), Volume 1*. Beijing: SDX Joint Publishing. [In Chinese].
- Qin, Baoqi, and Chao Meng. 2008. *Secret Societies and Qing Society* [UTF8gbns]. Tianjin: Tianjin Guji Chubanshe. [In Chinese].
- Qin, Baoqi. 2005. *China’s Underground Society, Volume 2: Secret Societies in the Late Qing* [UTF8gbns]. Beijing: Xueyuan Chubanshe. [In Chinese].
- Wakeman, Frederic, Jr. 1966. *Strangers at the Gate: Social Disorder in South China, 1839–1861*. Berkeley and Los Angeles: University of California Press.
- Wang, Yuanlin. 2012. *Internal and External Trade Economy: Lingnan Ports and Hinterlands, a Study of Overseas Transportation Relations*. Beijing: China Social Sciences Press. [In Chinese].
- Wu, Jincheng [Oh, Keum-sung]. 2018. *Spear and Shield: A Study of Jiangxi Society in the Ming and Qing Dynasties*. Translated by Ronggen Cui. Nanjing: Jiangsu People’s Press. [In Chinese].

- Yang, Ch'ing-k'un. 1975. "Some Preliminary Statistical Patterns of Mass Actions in Nineteenth-Century China." In *Conflict and Control in Late Imperial China*, edited by Carolyn Grant and Frederic E. Wakeman, 174–210. Berkeley: University of California Press.
- Yao, Xianhao. 1962. *Historical Materials on China's Modern Foreign Trade (1840–1895)*. Beijing: Zhonghua Book Company. [In Chinese].
- Zhang, Haiying. 2001. *Commodity Circulation and Market System in Jiangnan during the Ming and Qing Dynasties*. Shanghai: East China Normal University Press. [In Chinese].
- Zhang, Zhenguo. 2011. "Chong, Fan, Pi, Nan: A Preliminary Study of the Importance Rating System of Circuit, Prefecture, Department, and County in the Qing Dynasty." *Collected Essays on the Ming and Qing Dynasties* 11: 382–400.
- Zheng, Y., and T. Wu. 2026. "County-Level Population Dataset of Ancient China in 2, 742, 1102 and 1820 AD." *Scientific Data*. <https://doi.org/10.1038/s41597-026-07086-6>.

A Appendix

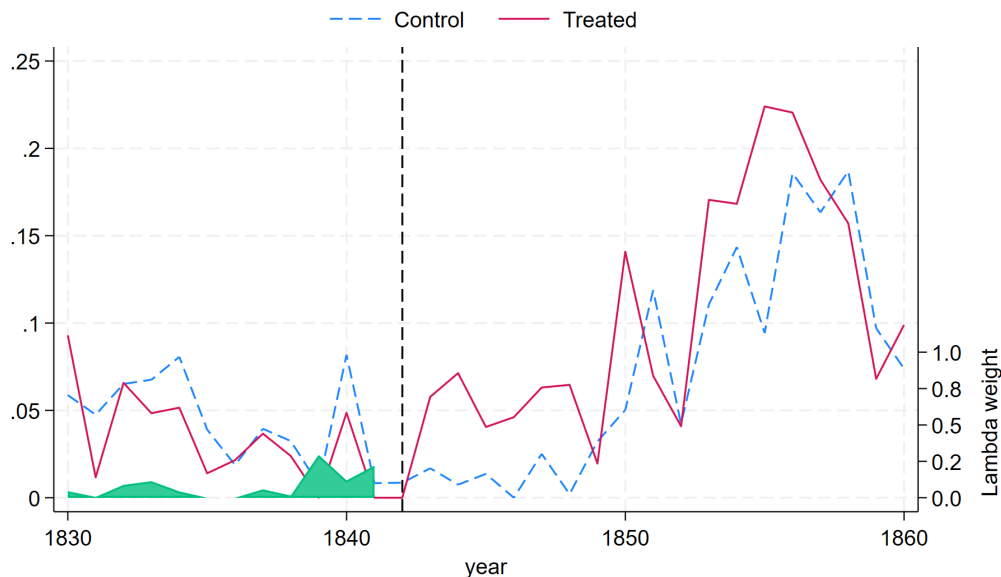


Figure A1: Synthetic Difference-in-Differences: Treated versus Synthetic Control Trajectories. The solid line is the corridor (treated) group and the dashed line is the synthetic control. The vertical dashed line marks the 1842 treatment date. The shaded area in the pre-treatment period and the right axis (*Lambda weight*) report the time weights selected by the synthetic difference-in-differences estimator.

Table A1: Robustness: Alternative Geographic Controls

	Rebellion occurrence (0/1)			
	(A1)	(A2)	(A3)	(A4)
Treated $\times$ Post-1842	0.1093*** (0.0184) [0.0218]	0.1080*** (0.0190) [0.0218]	0.1077*** (0.0185) [0.0218]	0.1055*** (0.0193) [0.0228]
Geographic control	Ruggedness	Wheat	Rice	Log(pop.)
County FE	Yes	Yes	Yes	Yes
Prov. $\times$ Year FE	Yes	Yes	Yes	Yes
Climate controls	Yes	Yes	Yes	Yes
March-route control	Yes	Yes	Yes	Yes
Mean of dependent variable	0.1412	0.1412	0.1412	0.1406
Number of counties	277	277	277	274
Observations	11,356	11,356	11,356	11,233
Adjusted R^2	0.329	0.329	0.329	0.329

Notes: The sample consists of 277 counties in four provinces (Anhui, Jiangxi, Fujian, and Guangdong) from 1820 to 1860, of which 51 are treated. The dependent variable is an indicator equal to one if at least one rebellion occurred in county c in year t . Each column re-estimates the kitchen-sink specification from Column (6) of Table 2, replacing the geographic control with an alternative: terrain ruggedness in Column (A1), wheat suitability in Column (A2), rice suitability in Column (A3), and log population in Column (A4). Each geographic control is interacted with Post-1842. All four candidates are highly correlated in our four-province sample ($|r| > 0.74$) and convey essentially the same information about plain versus upland terrain. All columns include county fixed effects, province $\times$ year fixed effects, climate controls (drought and flood indicators), and the march-route control (geodesic distance from each county centroid to the Taiping 1851 to 1853 military advance route, interacted with Post-1851). Standard errors clustered at the county level are reported in parentheses; Conley spatial-HAC standard errors with a 500 km spatial cutoff and 10-year serial correlation are reported in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Trade Shock and Rebellions: Positively Exposed Counties

	Rebellion occurrence (0/1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Treated $\times$ Post-1842	-0.0155 (0.0234) [0.0317]	0.0016 (0.0207) [0.0297]	0.0012 (0.0208) [0.0300]	0.0014 (0.0228) [0.0265]	-0.0032 (0.0207) [0.0280]	-0.0023 (0.0226) [0.0252]
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Prov. $\times$ Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	No	No	Yes
Geography controls	No	No	No	Yes	No	Yes
March-route control	No	No	No	No	Yes	Yes
Mean of dependent variable	0.0919	0.0919	0.0919	0.0919	0.0919	0.0919
Number of counties	150	150	150	150	150	150
Observations	6,150	6,150	6,150	6,150	6,150	6,150
Adjusted R^2	0.165	0.212	0.212	0.212	0.213	0.213

Notes: The sample consists of 150 counties in two provinces (Jiangsu and Zhejiang) from 1820 to 1860, of which 29 are treated. The dependent variable is an indicator equal to one if at least one rebellion occurred in county c in year t . Treated equals one for counties along the post-1842 maritime trade routes from the newly opened treaty ports; Post-1842 equals one in years 1842 and after. Post-1851 equals one in years 1851 and after. All columns include county fixed effects. Column (1) adds year fixed effects; Columns (2) to (6) replace year fixed effects with province $\times$ year fixed effects, absorbing all province-specific time trends. Climate controls comprise indicators for drought and flood (time-varying). The geography control is terrain ruggedness interacted with Post-1842. The march-route control is the geodesic distance from each county centroid to the Taiping 1851 to 1853 military advance route, interacted with Post-1851. Standard errors clustered at the county level are reported in parentheses; Conley spatial-HAC standard errors with a 500 km spatial cutoff and 10-year serial correlation are reported in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Trade Shock and Rebellions: Before Taiping Rebellion

	Rebellion occurrence (0/1), 1830–1850 only					
	(1)	(2)	(3)	(4)	(5)	(6)
Treated $\times$ Post-1842	0.0899*** (0.023)	0.1069*** (0.022)	0.1068*** (0.022)	0.1071*** (0.022)	0.1049*** (0.024)	0.1309*** (0.034)
climateanomaly				0.0086 (0.015)	0.0094 (0.015)	0.0122 (0.017)
drought_broad				0.0121 (0.016)	0.0092 (0.016)	-0.0026 (0.020)
flood_broad				0.0054 (0.011)	0.0044 (0.011)	0.0242* (0.013)
Constant	0.0744*** (0.002)	0.0730*** (0.002)	0.0788*** (0.002)	0.0717*** (0.009)	0.0311 (0.023)	0.0201 (0.025)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Province $\times$ Year FE	No	Yes	Yes	Yes	Yes	Yes
Crop failure	No	No	Yes	Yes	Yes	Yes
Climate anomaly	No	No	No	Yes	Yes	Yes
Drought	No	No	No	Yes	Yes	Yes
Flood	No	No	No	Yes	Yes	Yes
Geography $\times$ Post	No	No	No	No	Yes	Yes
Volatility (volatile R^2)	No	No	No	No	No	Yes
Mean of dependent variable	0.081	0.081	0.081	0.081	0.081	0.079
Number of observations	5817	5817	5817	5817	5817	4016
Number of counties	277	277	277	277	277	197
R-squared	0.138	0.240	0.241	0.241	0.242	0.238

Notes: The sample consists of 277 counties in four provinces (Anhui, Jiangxi, Fujian, and Guangdong) from 1820 to 1860, of which 51 are treated. The dependent variable is an indicator equal to one if at least one rebellion occurred in county c in year t . Treated $\times$ Post-1842 equals one for counties lying on the pre-1842 inland routes to Canton in years ≥ 1842 . All columns include county fixed effects. Column (1) includes year fixed effects; Columns (2) to (6) include province $\times$ year fixed effects and sequentially add controls for crop failure, climate anomaly, drought, and flood. Standard errors, clustered at the county level, in parentheses.

Table A4: Placebo Test: Positively-Exposed Sample (Jiangsu and Zhejiang)

	(1) Continuous	(2) Binary
TransitGain	-0.038 (0.038)	
Treated $\times$ Post1842		-0.009 (0.025)
Climate controls	Yes	Yes
Geography $\times$ Post1842	Yes	Yes
March-route $\times$ Post1851	Yes	Yes
County FE	Yes	Yes
Province $\times$ Year FE	Yes	Yes
Observations	4650	4650
Within-R ²	0.0064	0.0062

Notes: Standard errors clustered at the county level in parentheses. Estimated on the 150-county positively-exposed panel (Jiangsu, Zhejiang) over 1830–1860 ($N = 4,650$). Column (1) uses the continuous TransitGain = treatment $\times$ Shanghai_share_{*t*}, where Shanghai_share_{*t*} is the realized national share of overseas exports through Shanghai (tea and silk averaged). Column (2) uses Treated $\times$ Post1842. Treated = 1 for Jiangsu and Zhejiang counties on the new Shanghai inland feeder routes. Geography control: ruggedness $\times$ Post1842. March-route control: distance to Taiping 1851–1853 advance route $\times$ Post1851. Post1842 = 1 if year ≥ 1843 ; Post1851 = 1 if year ≥ 1851 . * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Continuous Timing Decomposition

	Occurrence	Count_Econ	Count_Poli
TransitLoss $\times$ Early	0.189*** (0.044)	0.118*** (0.037)	0.100*** (0.037)
TransitLoss $\times$ Late	0.078*** (0.028)	-0.034** (0.017)	0.107*** (0.032)
F-test (early=late)	4.943	21.395	0.031
F-test p-value	0.027	0.000	0.861
Climate controls	Yes	Yes	Yes
Geography $\times$ Post1842	Yes	Yes	Yes
March-route $\times$ Post1851	Yes	Yes	Yes
County FE	Yes	Yes	Yes
Province $\times$ Year FE	Yes	Yes	Yes
Observations	8587	8586	8561
Within-R ²	0.0071	0.0042	0.0123

Notes: Standard errors clustered at the county level in parentheses. Estimated on the 277-county negatively-exposed panel (Anhui, Fujian, Guangdong, Jiangxi) over 1830–1860. TransitLoss $\times$ Early = TransitLoss $\times \mathbf{1}(1842 \leq t \leq 1850)$; TransitLoss $\times$ Late = TransitLoss $\times \mathbf{1}(t \geq 1851)$. The F-test tests equality of the Early and Late coefficients. The Count_Poli sample is 26 observations smaller than the other two outcomes due to additional missingness on the political-violence count variable. Geography control: ruggedness $\times$ Post1842. March-route control: distance to Taiping 1851–1853 advance route $\times$ Post1851. Post1842 = 1 if year ≥ 1843 ; Post1851 = 1 if year ≥ 1851 . * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Binary Timing Decomposition

	Occurrence	Count_Econ	Count_Poli
Treated $\times$ Early	0.106*** (0.022)	0.098*** (0.018)	0.029* (0.016)
Treated $\times$ Late	0.089*** (0.022)	0.003 (0.012)	0.092*** (0.025)
F-test (early=late)	0.475	31.318	7.731
F-test p-value	0.491	0.000	0.006
Climate controls	Yes	Yes	Yes
Geography $\times$ Post1842	No	No	No
March-route $\times$ Post1851	No	No	No
County FE	Yes	Yes	Yes
Province $\times$ Year FE	Yes	Yes	Yes
Observations	8587	8586	8561
Within-R ²	0.0052	0.0070	0.0101

Notes: Standard errors clustered at the county level in parentheses. Estimated on the 277-county negatively-exposed panel (Anhui, Fujian, Guangdong, Jiangxi) over 1830–1860. Treated $\times$ Early = treatment $\times \mathbf{1}(1842 \leq t \leq 1850)$; Treated $\times$ Late = treatment $\times \mathbf{1}(t \geq 1851)$. The F-test tests equality of the Early and Late coefficients. The Count_Poli sample is 26 observations smaller than the other two outcomes due to additional missingness on the political-violence count variable. Table A8 retains the 4-control specification (climate variables only) to mirror the original paper’s binary-treatment Tables 3–4 and provide a comparison benchmark to the 6-control continuous specification in Table A7. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Endogeneity Robustness: Pre-Taiping Subsample

	Pre-Taiping (≤ 1851)
TransitLoss (standardized)	0.027*** (0.009)
Climate controls	Yes
Geography $\times$ Post1842	Yes
March-route $\times$ Post1851	Yes
County FE	Yes
Province $\times$ Year FE	Yes
Observations	6094
Within-R ²	0.0138

Notes: Standard error clustered at the county level in parentheses. Sample restricted to year ≤ 1851 ($N = 6,094$) to eliminate any post-1851 Canton-share variation potentially driven by the Taiping rebellion. TransitLoss (standardized) is the TransitLoss variable rescaled to mean zero and standard deviation one over the full 277-county sample. Geography control: ruggedness $\times$ Post1842. March-route control: distance to Taiping 1851–1853 advance route $\times$ Post1851. Post1842 = 1 if year ≥ 1843 ; Post1851 = 1 if year ≥ 1851 . * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Covariate Balance Before and After Propensity-Score Matching

Covariate	Sample	Mean		% bias	<i>t</i> -test	
		Treated	Control		<i>t</i>	<i>p</i>
Terrain ruggedness	Unmatched	147.9506	166.4183	−19.3	−1.25	0.214
	Matched	159.8991	162.6229	−2.8	−0.13	0.899
County area	Unmatched	2,267.62	2,199.2	6.5	0.42	0.678
	Matched	2,190.5	2,194.21	−0.4	−0.01	0.988
1820 population	Unmatched	521,732	296,751	54.8	4.63	0.000
	Matched	326,729	369,952	−10.5	−0.76	0.447
Market town density	Unmatched	.0019791	.0018305	13.3	0.80	0.426
	Matched	.0019171	.0020512	−12.0	−0.52	0.602
Rice suitability	Unmatched	3,011.03	2,685.81	28.7	1.83	0.068
	Matched	2,902.99	2,970.68	−6.0	−0.25	0.802
Wheat suitability	Unmatched	2,276.66	2,454.08	−19.1	−1.17	0.244
	Matched	2,255.53	2,362.67	−11.5	−0.50	0.615
Flood frequency (1830–41)	Unmatched	.5620915	.4816891	81.2	4.96	0.000
	Matched	.5595238	.5674603	−8.0	−0.39	0.698
Drought frequency (1830–41)	Unmatched	.2091503	.2679372	−44.1	−2.71	0.007
	Matched	.2063492	.218254	−8.9	−0.46	0.645
Pre-shock rebellion frequency (1830–41)	Unmatched	.1094771	.1008969	8.3	0.54	0.588
	Matched	.1031746	.1210317	−17.2	−0.75	0.458
Mean absolute % bias		30.6 → 8.6				
Pseudo- R^2		0.218 → 0.051				
Joint LR test (<i>p</i> -value)		0.000 → 0.745				

The propensity score is from a logit of the Canton-route treatment on the nine covariates above, estimated on the 274-county cross-section (3 of 277 baseline counties dropped for missing covariates). Matching is 1-to-1 nearest-neighbour, caliper 0.05, no replacement: 42 of 51 treated counties are matched to 42 control counties (84 counties total).

For each covariate, % bias is the standardized percentage difference in means between treated and control; the *t*-test is for the treated–control difference. *Unmatched* is the raw 274-county cross-section; *Matched* is the 84-county matched sample. Computed with the `psctest` standardized-bias formulas (unmatched-group variances in the denominator before and after).

Bottom panel: mean absolute standardized % bias across covariates, the propensity-score logit pseudo- R^2 , and the *p*-value of the likelihood-ratio test of joint covariate orthogonality to treatment, each reported as Unmatched → Matched.

Table A9: Propensity-Score-Matching Robustness: Baseline Canton-Route DID on the Matched Sample

	Full sample	Matched sample
Treated $\times$ Post1842	0.0831*** (0.0176)	0.0795*** (0.0261)
County FE	Yes	Yes
Province $\times$ Year FE	Yes	Yes
Four controls	Yes	Yes
Observations	8,586	2,603
Counties (clusters)	277	84

Dependent variable: `occurrence`, a binary indicator for any rebellion event in the county-year; panel 1830–1860, `post1842` = 1(year $\geq$ 1842). Column (1) is the full 277-county baseline sample; column (2) restricts to the 84 counties retained by 1-to-1 nearest-neighbour propensity-score matching (caliper 0.05, no replacement; 42 of 51 treated counties matched to 42 controls).

All columns include county fixed effects, province $\times$ year fixed effects, and the four baseline controls (`drought_broad`, `flood_broad`, `ruggedness` $\times$ `Post1842`, `dist_march_route` $\times$ `Post1851`).

Standard errors clustered by county in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Changes-in-Changes: Quantile Treatment Effects (rebellion-intensity outcome, incident count)

Quantile	QTE	Bootstrap SE	95% CI
Overall (ATE)	0.1532***	0.0320	[0.0905, 0.2159]
q5 [†]	0.0459	0.0196	[0.0076, 0.0843]
q10 [†]	0.0624	0.0225	[0.0183, 0.1065]
q25	0.0205**	0.0092	[0.0026, 0.0384]
q50	0.0191***	0.0072	[0.0050, 0.0331]
q75	0.0178***	0.0042	[0.0097, 0.0259]
q90	0.1628***	0.0230	[0.1177, 0.2080]
q95	0.2486***	0.0506	[0.1494, 0.3479]

Changes-in-changes estimates follow Athey and Imbens (2006), implemented via Krainer’s `cic` command; the Melly–Blaise variant used by Cao and Chen (2022) was unavailable, though the estimand is identical. Because the rebellion-intensity outcome has a large mass at zero (84.5% of county-years), quantile treatment effects below the 25th percentile rest on sparse information and should be read with caution; estimates at and above the 25th percentile are significant, with the largest effects in the upper tail.

Standard errors from 100 bootstrap replications. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

† Quantile treatment effects at q5 and q10 fall within the zero-mass region of the outcome distribution (84.5% of county-years are zero) and are reported for completeness but not interpreted; the identified and interpretable range is q25 and above.

Table A11: Robustness of Three Estimators to Outcome Coding: Occurrence vs. Incident Count

	Occurrence (0/1)	Incident count
<i>Panel A. Baseline two-way FE DID, asinh outcome</i>		
Treated×Post (county + prov×year FE)	0.0327*** (0.0102)	0.0417*** (0.0116)
Treated×Post (+ four controls)	0.0257** (0.0102)	0.0342*** (0.0115)
<i>Panel B. Changes-in-Changes</i>		
Overall ATE	0.0742*** (0.0175)	0.1532*** (0.0320)
QTE q25	0.0086 (0.0093)	0.0205** (0.0092)
QTE q50	0.0056 (0.0062)	0.0191*** (0.0072)
QTE q90	0.1752*** (0.0214)	0.1628*** (0.0230)
<i>Panel C. Synthetic DID (ATT; pre-period RMSPE in brackets)</i>		
v1: all donors	0.0382*** (0.0099) [0.0216]	0.0426*** (0.0102) [0.0287]
v2: buffer 52 km	0.0422*** (0.0089) [0.0280]	0.0428*** (0.0100) [0.0374]

Results are reported under both outcome codings: a binary occurrence indicator and an incident count, each normalized by county population and inverse-hyperbolic-sine transformed. Sign, magnitude, and significance are stable across codings.

Standard errors in parentheses; Synthetic DID standard errors are placebo-based. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Synthetic Difference-in-Differences: ATT on Rebellion Occurrence

	ATT	Placebo SE	Bootstrap SE	Jackknife SE	p	Treated/Control
v1: all donors (222)	0.0382***	0.0099	0.0099	0.0102	<0.001	51/222
v2: buffer 52 km (167)	0.0422***	0.0089	0.0107	0.0108	<0.001	51/167

Synthetic DID follows Arkhangelsky et al. (2021). Given the short pre-treatment window (1830–1841), the synthetic control fit is imperfect (pre-period RMSPE ≈ 0.022 , about 0.57 of the estimated ATT); we therefore report it as corroborating rather than primary inference. Synthetic DID is preferred over the classical synthetic control method precisely because it is more robust to pre-treatment panels.

v1 uses all 222 control counties as donors (N=8,463); v2 drops 55 controls within 52 km of the corridor, leaving 167 donors. Treatment is block-assigned at 1842. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Commodity Composition: Tea vs Silk Corridor Exposure

	(1)	(2)	(3)	(4)	(5)	(6)
FLR tea	0.619*** (0.154)	0.647*** (0.157)	0.647*** (0.157)	0.611*** (0.153)	0.611*** (0.153)	0.611*** (0.153)
FLR silk	0.003 (0.036)	0.002 (0.038)	0.002 (0.038)	0.016 (0.038)	0.016 (0.038)	0.016 (0.038)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Province×Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	Yes	Yes	Yes
Geography control	No	No	No	Yes	Yes	Yes
March-route control	No	No	No	No	Yes	Yes
p -value: FLR tea = FLR silk	0.000	0.000	0.000	0.001	0.001	0.001
Mean of dependent variable	0.0815	0.0815	0.0815	0.0815	0.0815	0.0815
Number of counties	277	277	277	277	277	277
Observations	5817	5817	5817	5817	5817	5817
Adjusted R^2	0.093	0.190	0.189	0.190	0.190	0.190

Notes: Standard errors clustered at the county level (OID) are in parentheses. The exposure measure is $FLR_{ct}^k = \mathbf{1}\{\text{commodity } k \text{ transits the corridor of county } c\} \times (1 - s_{\text{Canton},t}^k)$, where $s_{\text{Canton},t}^k$ is Canton's share of national exports of commodity k in year t . Corridor-commodity assignment follows Zhang (2001), Yao (1962), and Fortune (1999): tea corridors comprise the Gan-river trunk together with the Wannan and Wuyi feeder branches, and silk corridors comprise the trunk and the lower-Yangzi silk approach. The estimation window is restricted to 1830–1850 to avoid Taiping-era contamination of post-1851 port shares. The row “ p -value: FLR tea = FLR silk” reports an F -test of the equality of the two coefficients. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: Commodity Composition, Full Window (1830–1860)

	(1)	(2)	(3)	(4)	(5)	(6)
FLR tea	0.121** (0.061)	0.092** (0.047)	0.092** (0.047)	0.084* (0.048)	0.077 (0.047)	0.077 (0.047)
FLR silk	0.048 (0.038)	0.023 (0.029)	0.024 (0.029)	0.033 (0.029)	0.026 (0.029)	0.026 (0.029)
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Province×Year FE	No	Yes	Yes	Yes	Yes	Yes
Climate controls	No	No	Yes	Yes	Yes	Yes
Geography control	No	No	No	Yes	Yes	Yes
March-route control	No	No	No	No	Yes	Yes
p-value: FLR_tea = FLR_silk	0.423	0.316	0.320	0.476	0.458	0.458
Mean of dependent variable	0.1559	0.1559	0.1559	0.1559	0.1559	0.1559
Number of counties	277	277	277	277	277	277
Observations	8587	8587	8587	8587	8587	8587
Adjusted R^2	0.191	0.347	0.348	0.348	0.349	0.349

Standard errors in parentheses

$FLR^k_{ct} = 1commodityktransitscountyc'scorridorx(1 - s^k_{Canton}, t);$

corridor-commodity assignment follows Zhang (2001), Yao (1962), Fortune (1999);

tea corridors = trunk_full + branch_wannan + branch_wuyi; silk corridors = trunk_full + trunk_silk;

post-1851 port shares are disturbed by the Taiping rebellion – this full-window table is a window-sensitivity diagnostic only; clustered SE at county (OID).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$